

Chapter 1

Discrete Time Signals and Systems.

1.1. What are Signals?

Signals surround us and play an important role in our lives. Basically, Signals are carriers of information. When you speak into a mobile phone in Kathmandu, when an ECG machine monitors a heartbeat at Bir Hospital, or when a seismometer detects earth tremors in Gorkha, you are dealing with signals. Some examples of signals that we encounter frequently in our daily life are speech, music, picture, and video signals. **A signal is a function of independent variables such as time and space, which can be one or more.** For example, speech and music signals represent air pressure as a function of time at a point in space. A black-and-white picture is a representation of light intensity as a function of two spatial coordinates. The video signal in television consists of a sequence of images, called frames, and is a function of three variables: two spatial coordinates and time.

The majority of signals we encounter are generated naturally. Most natural signals exist in a continuous-time (analog) form. A signal can, however, also be generated artificially or through computer simulation. **Signal processing**

refers to methods used to extract information from signals. The specific method used to extract the information depends on the type of signal and the nature of the information it contains. Therefore, signal processing is essentially the process of mathematically representing a signal and applying an algorithmic operation to it in order to extract the information contained in the signal. However, modern computers, microcontrollers (like the Arduino used in your labs), and smartphones cannot process continuous waveforms directly. **This course is concerned with the discrete-time representation of signals and their discrete-time processing.**

This chapter provides an overview of signals and signal processing techniques. First, the classification of signals and the mathematical characterization of the signal are discussed. The sort of information carried by a few common signals is then defined in depth, along with the discussion of those signals. Then, through the use of examples, an overview of a few frequently used signal processing operations is given. Furthermore, we define basic discrete-time sequences, classify them based on energy and power, explore the unique quirk of periodicity in the digital domain, and learn to manipulate the time index. We will then introduce Fourier analysis to view signals in the frequency domain. Finally, we define systems and discuss the critical process of sampling, i.e., the bridge that connects the real analog world to our digital calculations.

1.2. Classification of signals

Before we focus exclusively on discrete-time signals, it is useful to understand the broader taxonomy of signals. Signals can be categorized based on the nature of their independent variables, their values, and their predictability.

1. Based on the Independent Variable

- Continuous-Time Signals: These types of signals are defined for all values of time t . Denoted as $x(t)$. Examples include the voltage output from a microphone or the varying air pressure of a Bansuri flute.
- Discrete-Time Signals: These signals are defined only at integer values of the index n . Denoted as $x[n]$. This is the primary focus of this textbook.
- Digital Signals: A discrete-time signal whose amplitude is also quantized to a finite set of values (e.g., 16-bit integers in a .wav file) is called a digital signal.

2. Based on Origin and Generation

- Natural Signals: Natural signals are generated by physical phenomena. These are inherently continuous-time and analog until digitized. Common examples include:
 - Audio Signals: Speech, music, environmental sounds (e.g., bird calls, jackie barking, river flow).
 - Biological Signals: ECG (heart electrical activity), EEG (brain waves).
 - Geophysical Signals: Seismic waves recorded at the National Seismological Centre.
- Synthetic Signals: These are generated artificially by computers or function generators (e.g., a test tone, a digital clock pulse, a simulated sine wave).

3. Based on Dimensionality

- One-Dimensional (1-D) Signal: A function of a single independent variable. Example: Speech $x(t)$ varies only with time.

- Multidimensional (M-D) Signal: A function of two or more independent variables.
 - 2-D Signal: A grayscale image $I(x, y)$ depends on two spatial coordinates (horizontal and vertical).
 - 3-D Signal: A color video depends on two spatial coordinates, one time coordinate, and three color channels (RGB).

4. Based on Value Characteristics

- Scalar versus Vector Signals:
 - Scalar Signal: A single value at each instant (e.g., temperature reading, single-channel microphone input).
 - Vector (Multichannel) Signal: Multiple components at each instant. Examples include stereo audio (left/right channels), RGB color images, or 3-axis accelerometer data used in smartphone orientation sensing.
- Real versus Complex Signals:
 - Real-Valued: $x[n] \in \mathbb{R}$. Most physical measurements are real.
 - Complex-Valued: $x[n] \in \mathbb{C}$. Used heavily in Fourier analysis (Section 1.5 and 1.6) and telecommunications (IQ signals) to represent both magnitude and phase compactly.

5. Based on Predictability

- Deterministic Signals: The deterministic signal can be described exactly by a mathematical formula or a fixed rule. The future values are known precisely. Example: $x[n] = \cos(0.2\pi n)$.
- Random (Stochastic) Signals: These signals cannot be predicted exactly in advance. They are described using statistical measures (mean,

variance, autocorrelation). Examples include thermal noise in electronic circuits or a random Nepali speech signal.

Note: While random signals are crucial in advanced DSP, the core analysis tools in this introductory chapter focus on deterministic discrete-time signals.

1.3. Discrete-time signals (Sequences)

Mathematically, a discrete-time signal is defined as **sequences of numbers**. A sequence of numbers x , in which the n^{th} number in the sequence is denoted $x[n]$, is formally written as

$$x = \{x[n]\}, \quad -\infty < n < \infty, \quad (1.1)$$

where,

n is an integer.

If a discrete-time signal is written as a sequence of number inside curly braces, the location of sample value at time index $n = 0$ is indicated by $\uparrow$ under the sample value. For example,

$$x = \{x[n]\} = \{\dots, 3.4, 0.95, 2.15, 1.3, 2.94, -2.3, -3.5, 5.0, 7.4, \dots\} \quad (1.2)$$

$\uparrow$

For the above signal, $x[-1] = 2.94$, $x[0] = -2.3$, and so on. The graphical representation of the above sequence is shown in Figure (1.1). **The length of the sequence is the number of samples in the sequence.** The discrete-time signal may be finite-length sequence or infinite-length sequence. An infinite

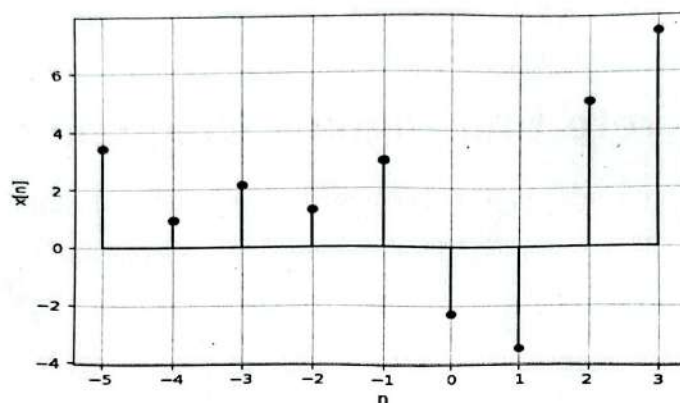


Figure 1.1: Graphical representation of $x[n]$

length sequence is defined for $-\infty < n < \infty$, whereas finite-length sequence is defined to $N1 \leq n \leq N2$, where

$$-\infty < N1,$$

$$N2 < \infty, \text{ and}$$

$$N1 < N2.$$

The length of finite-length sequence is $N = N2 - N1 + 1$.

1.3.1. Basic Operation on discrete time signal

In DSP, operations on discrete-time signals can be broadly classified into two categories based on which variable is being modified:

1. Transformation of dependent variable, and
2. Transformation of independent variable

1.3.1.1. Transformation of Dependent variable

This involves changing the signal's amplitude, i.e., value.

- **Addition:** Let $x[n]$ and $y[n]$ be two sequences. The first basic operation is the addition by which a new sequence $w_1[n]$ is obtained by adding the sample values of two sequences, $x[n]$ and $y[n]$:

$$w_1[n] = x[n] + y[n]. \quad (1.3)$$

The device implementing the addition operation is called an *adder*. Example: Mixing two audio signals.

- **Multiplication:** The second basic operation is multiplication, by which a new sequence $w_2[n]$ is obtained by the product of the sample values of these two sequences at each time instant:

$$w_2[n] = x[n] \cdot y[n]. \quad (1.4)$$

The device implementing the multiplication operation is called a *multiplier*. Example: AM/FM modulation in communication.

- **Amplitude Scaling/Scalar Multiplication:** The third basic operation is the scalar multiplication, where a new sequence is generated by multiplying each sample of a sequence $x[n]$ by a scalar A :

$$w_3[n] = Ax[n] \quad (1.5)$$

where,

$A \in \mathbb{R}$.

Example: Increasing/decreasing audio volume.

1.3.1.2. Transformation of Independent variable

This involves changing the time/index, i.e., n .

- **Time Shifting (Delay/Advancing):** Let $x[n]$ be a discrete-time signal, then the time-shifting operation is defined as:

$$w_4[n] = x[n - N] \quad (1.6)$$

where, $N \in \mathbb{Z}$

- if $N > 0$, we say the operation is delayed, and
- if $N < 0$, we say the operation is advanced.

Figure 1.2 shows an example of time delay operation; similarly, Figure 1.3 shows an example of time advance operation. The device implementing shifting operation by one sample is called an *unit delay*.

- **Time Reversal/Folding:** The time-reversal is also called folding operation. Let us consider a discrete time signal $x[n]$ then the time folded signal $y[n]$ is defined as:

$$y[n] = x[-n]$$

Note: The folding is usually done at $n = 0$, but the folding can also be done at any value of n . Figure 1.4 show an example of time reversal of unit step discrete time signal. Example: Playing a signal backward.

- **Time Scaling:** The time scaling of discrete time signal $x[n]$ can be either compression or expansion which are defined as:

$$y[n] = x[kn], \text{ for compression}$$

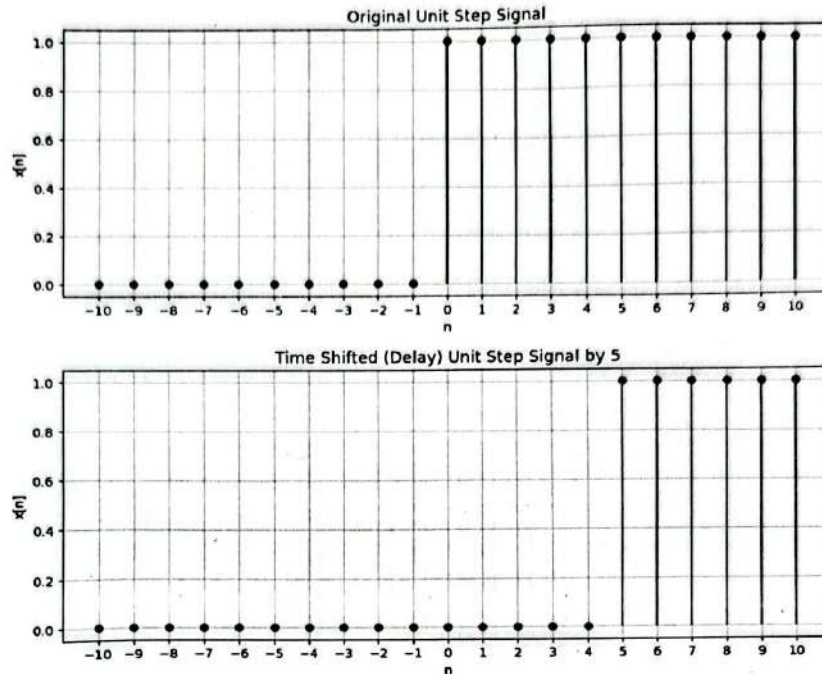


Figure 1.2: Unit step signal delayed by 5 units

and,

$$y[n] = x\left[\frac{n}{k}\right], \text{ for expansion}$$

where,

k is called the scaling factor, and $k > 1$.

Example 1.1 Compression:

Let us consider: $x[n] = \{0, 1, 1, 1, 1, 0, 0.7, 0.7, 0.7, 0.7, 0\}$ defined for $n = 0$ to 10 as shown in figure 1.5 Let $k = 2$, i.e., compression or downsampling by 2, we get $x[2n] = \{0, 1, 1, 0.7, 0.7, 0\}$ for $n = 0$ to 5 as shown in figure 1.5.

Note: In compression, we keep every k -th sample and discard the rest.

Example 1.2 Expansion:

Let us consider: $x[n] = \{0, 1, 1, 1, 1, 0, 0.7, 0.7, 0.7, 0.7, 0\}$ defined for $n = 0$ to 10 as shown in figure 1.5 Let $k = 2$, i.e. expansion or upsampling by 2,

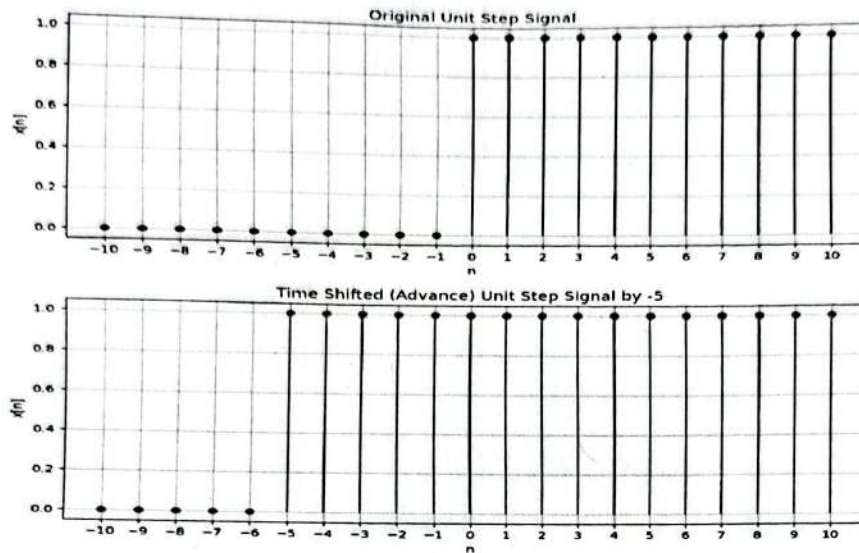


Figure 1.3: Unit step signal advanced by 5

we get $x[\frac{n}{2}] = \{0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 0.7, 0, 0.7, 0, 0.7, 0, 0.7, 0, 0, 0\}$ for $n = 0$ to 21 as shown in figure 1.6

Note: In Expansion, we insert k zero after every sample and discard.

Figure 1.7, and 1.8 shows another example for signal compression and expansion, respectively.

1.3.2. Basic discrete-time signals (sequences)

1. Unit sample or Impulse sequence:

The unit sample sequence is defined as:

$$\delta[n] = \begin{cases} 1 & \text{for } n = 0, \\ 0 & \text{for } n \neq 0. \end{cases} \quad (1.7)$$

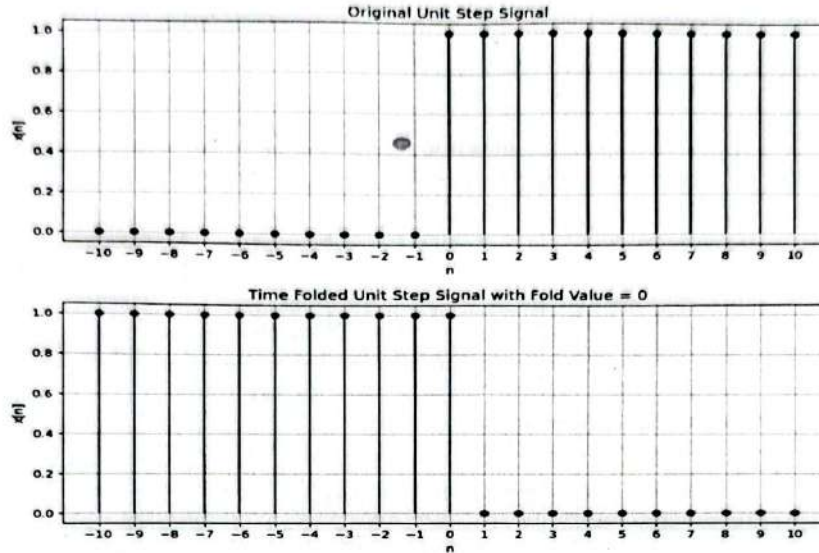


Figure 1.4: Time reversal of unit step signal

Unit sample sequence are also called impulse sequence or simply as an impulse. The unit sample sequence plays the same role for discrete-time signals and systems that the unit impulse function (Dirac delta function) does for continuous-time signals and systems. The graphical representation of the impulse sequence is shown in Figure (1.9).

- **Any arbitrary sequence can be represented as a sum of scaled and delayed impulses:** Let us consider an arbitrary signal:

$$x = \{x[n]\} = \{-3, 1.6, -2.4, 1.5, -1.9\} \quad (1.8)$$

↑

Each sample of above arbitrary sequence and be decomposed as weighted and shifted impulses as shown in Figure 1.10 Mathematically, sequence

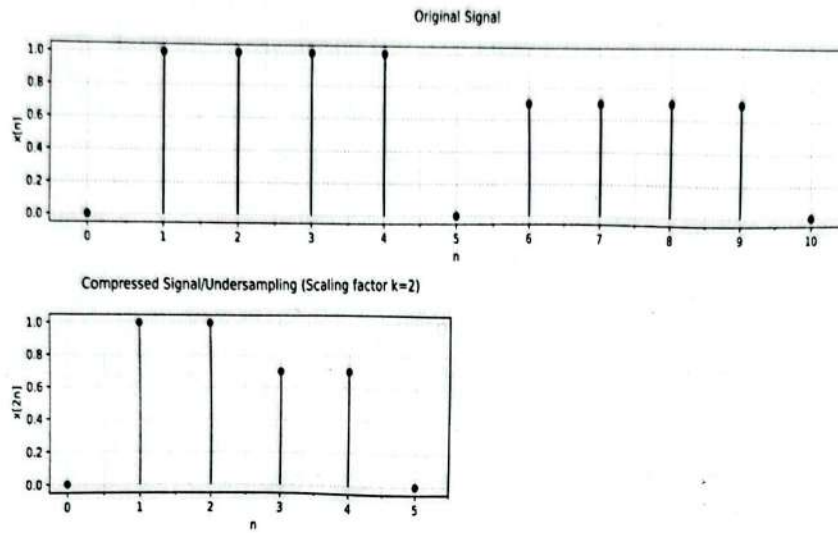


Figure 1.5: Original and compression signal with scaling factor $k = 2$

$x[n]$ can be written as:

$$x[n] = x[-2] \cdot \delta[n+2] + x[-1] \cdot \delta[n+1] + x[0] \cdot \delta[n] + x[1] \cdot \delta[n-1] + x[2] \cdot \delta[n-2]$$

In general for arbitrary sequence we have,

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta(n-k) \quad (1.9)$$

2. Unit step sequence:

The unit step sequence is defined as:

$$u[n] = \begin{cases} 1 & \text{for } n \geq 0, \\ 0 & \text{for } n < 0. \end{cases} \quad (1.10)$$

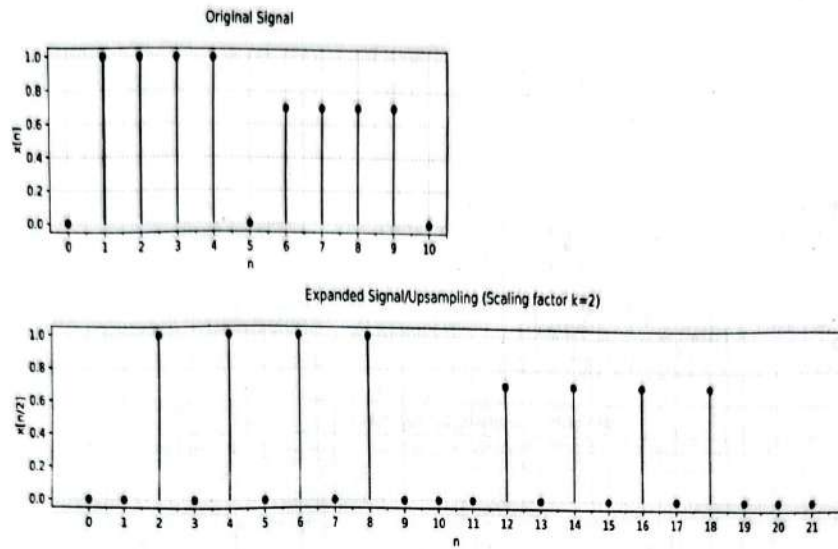


Figure 1.6: Original and Expansion signal with scaling factor $k = 2$

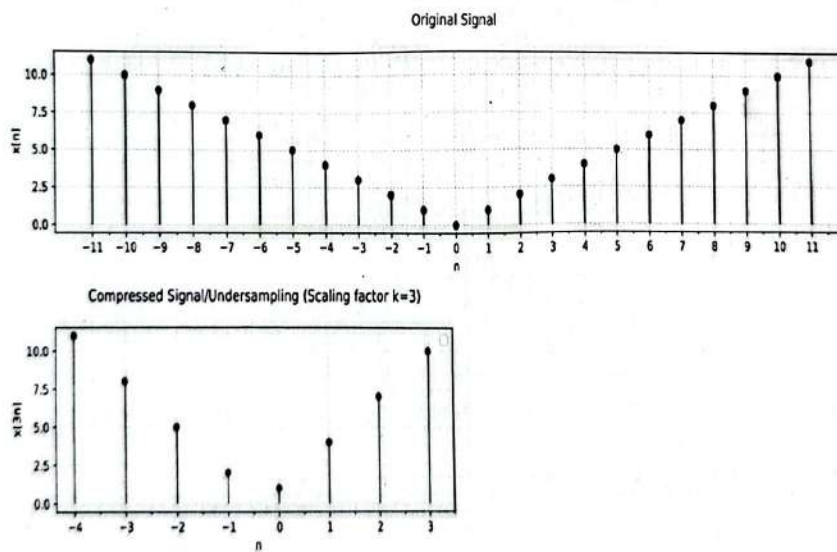


Figure 1.7: Original and Compression signal with scaling factor $k = 2$

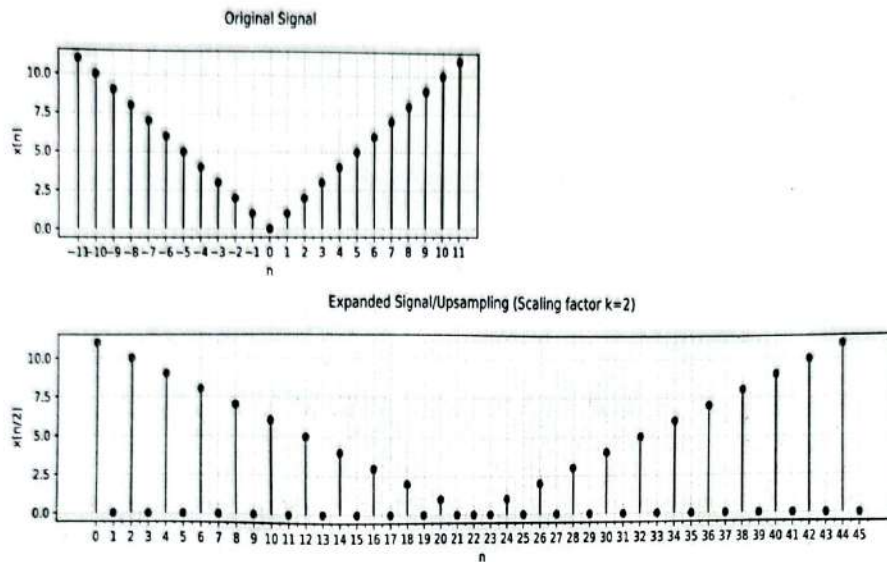


Figure 1.8: Original and Expansion signal with scaling factor $k = 2$

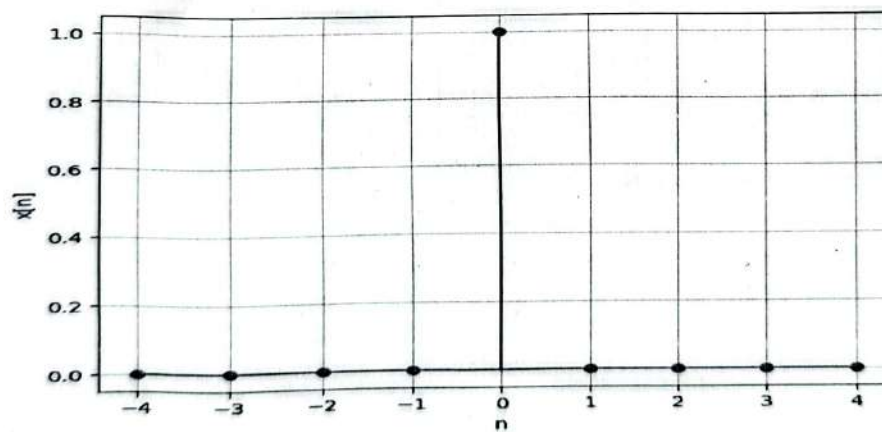


Figure 1.9: Unit Sample sequence

The unit step is related to the unit impulse as:

$$u[n] = \sum_{k=0}^n \delta[n - k]. \quad (1.11)$$

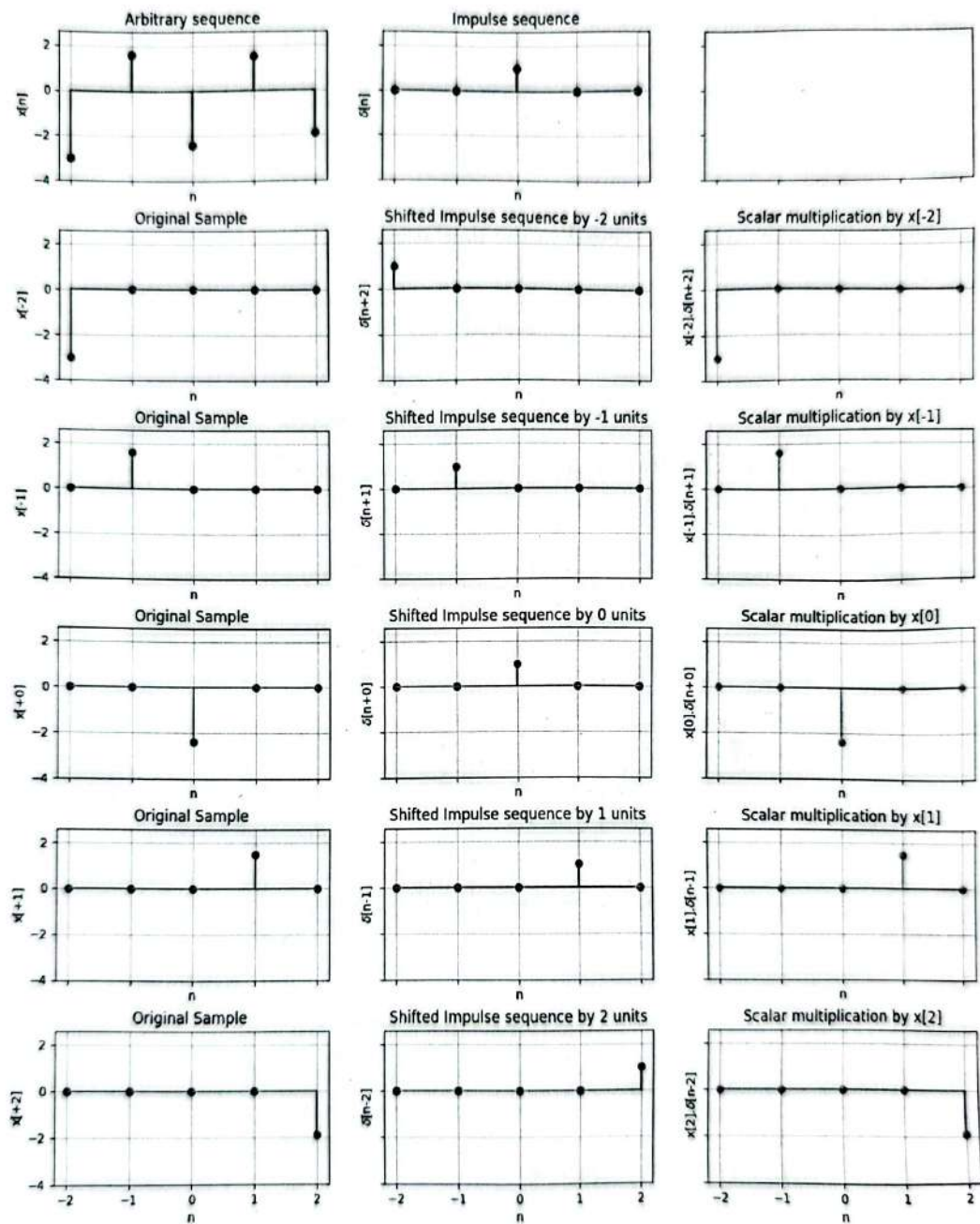


Figure 1.10: Representation of each sample of arbitrary sequence as scaled and shifted impulses.

Note: To show the relationship, plot the time-shifted unit impulse sequence.

The unit sample is related to the unit step as:

$$\delta[n] = u[n] - u[n - 1]. \quad (1.12)$$

Note: To show the relationship, plot the sequences, $u[n]$ and $u[n - 1]$.

The graphical representation of the unit step sequence is shown in Fig. (1.11).

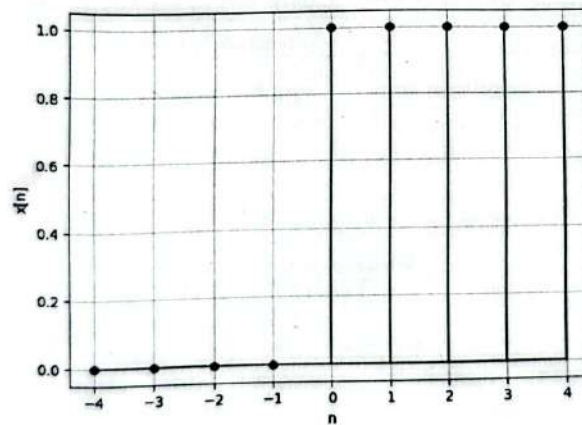


Figure 1.11: Unit Sample sequence

3. Ramp sequence:

A ramp sequence is defined as:

$$r[n] = \begin{cases} n & \text{for } n \geq 0, \\ 0 & \text{for } n < 0. \end{cases} \quad (1.13)$$

The graphical representation of the unit ramp sequence is shown in Fig.

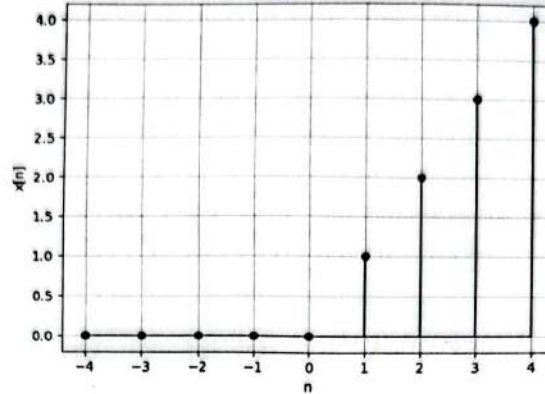


Figure 1.12: Ramp sequence

(1.12).

4. Signum sequence:

A Signum sequence is defined as:

$$sgn[n] = \begin{cases} 1 & \text{for } n > 0, \\ 0 & \text{for } n = 0, \\ -1 & \text{for } n < 0 \end{cases} \quad (1.14)$$

The graphical representation of the signum sequence is shown in Fig.

(1.13).

5. Sinusoidal sequence:

A discrete-time sinusoidal sequence is defined as:

$$x[n] = A \cos(\omega_0 n + \phi), \quad -\infty < n < \infty \quad (1.15)$$

where,

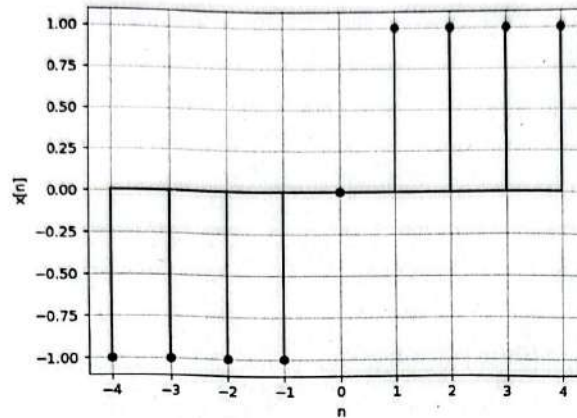


Figure 1.13: A Signum sequence.

A, ω_o , and $\phi \in \mathcal{R}$ and are called Amplitude, angular frequency and phase of the sinusoidal sequence,

$$\omega_o = 2\pi f_o$$

6. **Exponential sequence** An exponential sequence is defined as:

$$x[n] = A\alpha^n \quad (1.16)$$

• **A and α are real numbers**, then the sequence is real exponential sequence:

- If $0 < \alpha < 1$ and A is positive then the sequence values are positive and decrease with increasing n ,
- if $-1 < \alpha < 0$ the sequence values alternate in sign but again decrease in magnitude with increasing n ,
- If $\alpha > 1$, then the sequence grows exponentially in magnitude as n increases,
- If $\alpha < -1$, then the sequence decays exponentially in magnitude as

n increases.

Fig. (1.14) shows the plot of real exponential sequence for various values of α .

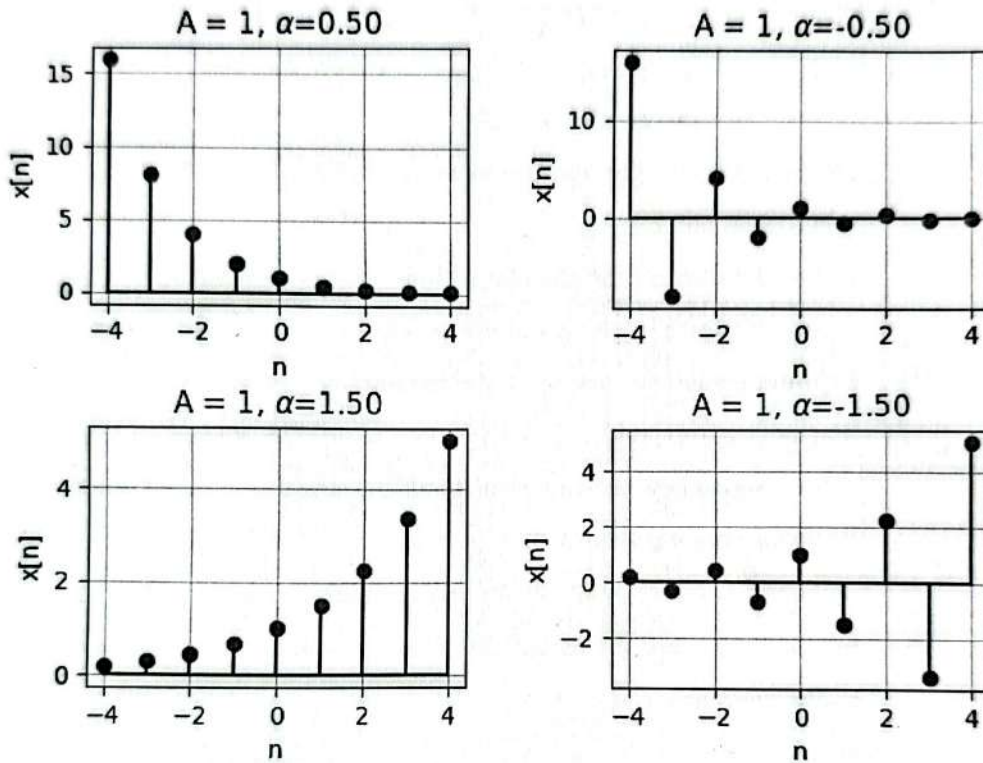


Figure 1.14: Plot of real exponential sequence for various values of α .

- A and α are complex numbers, then the sequence is complex exponential sequence. Expressing A and α as:

$$A = |A|e^{j\phi} \text{ and } \alpha = e^{\sigma_0 + j\omega_0} \text{ in equation (1.16), we get:}$$

$$\begin{aligned}
x[n] &= |A|e^{j\phi}e^{(\sigma_0+j\omega_0)n} \\
&= |A|e^{\sigma_0 n}e^{j(\omega_0 n + \phi)} \\
&= |A|e^{\sigma_0 n}\cos(\omega_0 n + \phi) + |A|e^{\sigma_0 n}\sin(\omega_0 n + \phi)
\end{aligned}$$

Let us understand the above equation with various plots for different values of ω_0 and σ_0 :

Figure (1.15) shows the plots for various value of ω_0 when $\sigma_0 = 0$ and $n > 0$. Notice how the amplitude of all the sequences are **neither exponentially increasing or decreasing** as $\sigma_0 = 0$.

Figure (1.16) shows the plots for various value of ω_0 when $\sigma_0 > 0$ and $n > 0$. Notice how the amplitude of all the sequences are **exponentially increasing** as $\sigma_0 > 0$.

Figure (1.17) shows the plots for various value of ω_0 when $\sigma_0 < 0$ and $n > 0$. Notice how the amplitude of all the sequences are **exponentially decreasing** as $\sigma_0 < 0$.

1.3.3. Precedence Rule for Discrete-Time Signals

Having understood the basic discrete-time signal and basic operations on discrete-time signals, we are now ready to perform complex operations on discrete-time signals. For a discrete-time signal $x[n]$, the combined operation

$$y[n] = x[an + b], \quad (1.17)$$

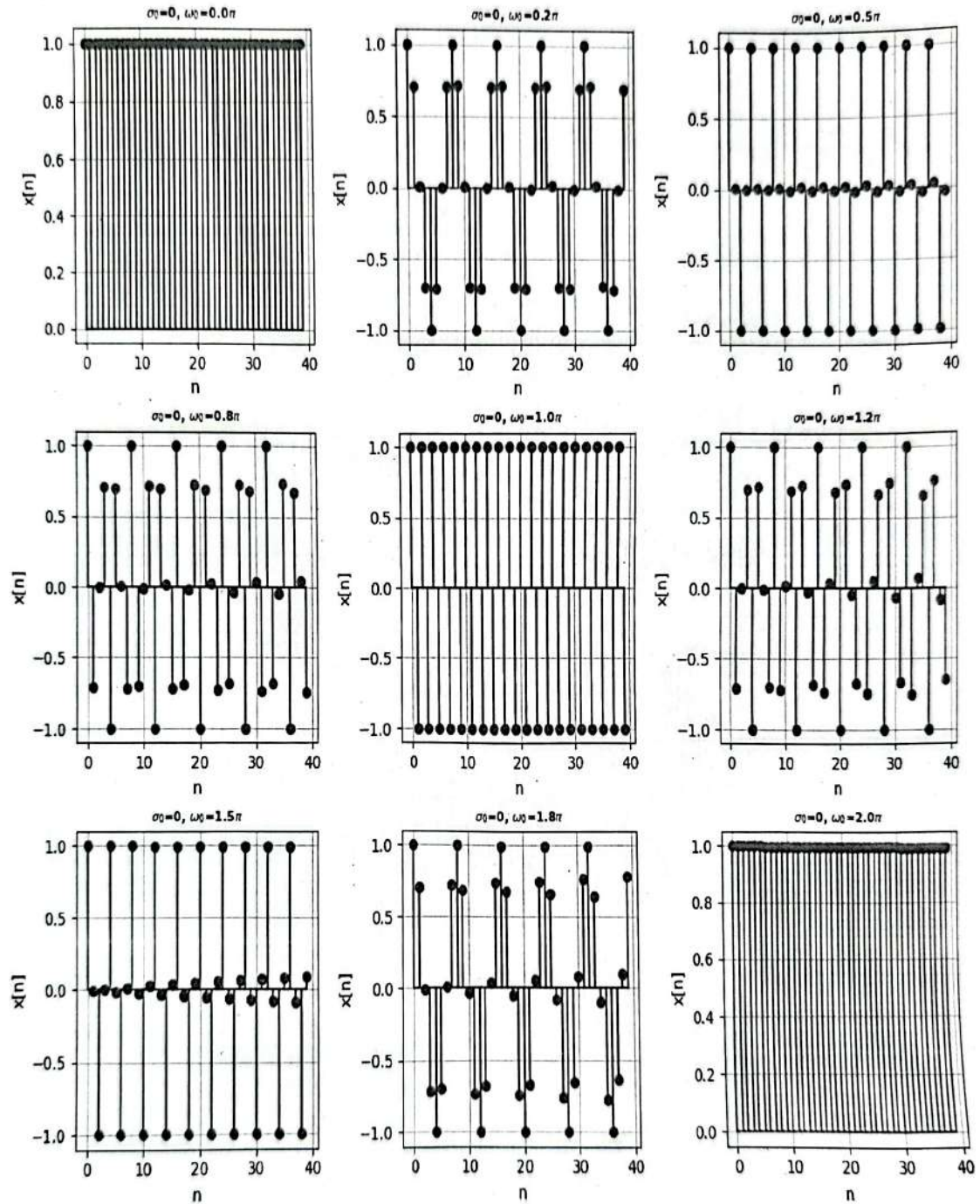


Figure 1.15: Plot of complex exponential sequence for various values of ω_0 , $\sigma_0 = 0$ and $n > 0$.

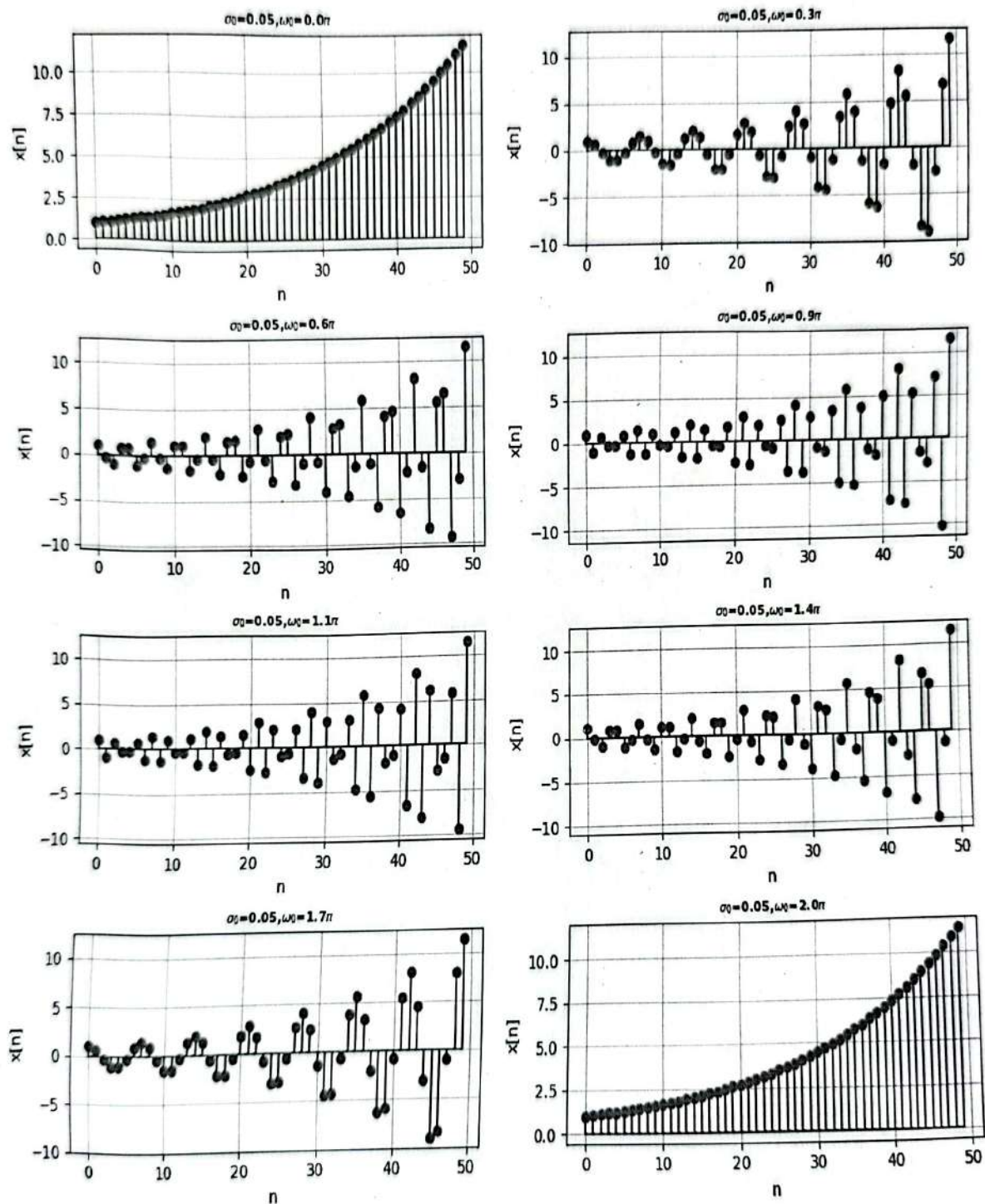


Figure 1.16: Plot of complex exponential sequence for various values of ω_0 , $\sigma_0 > 0$ and $n > 0$.

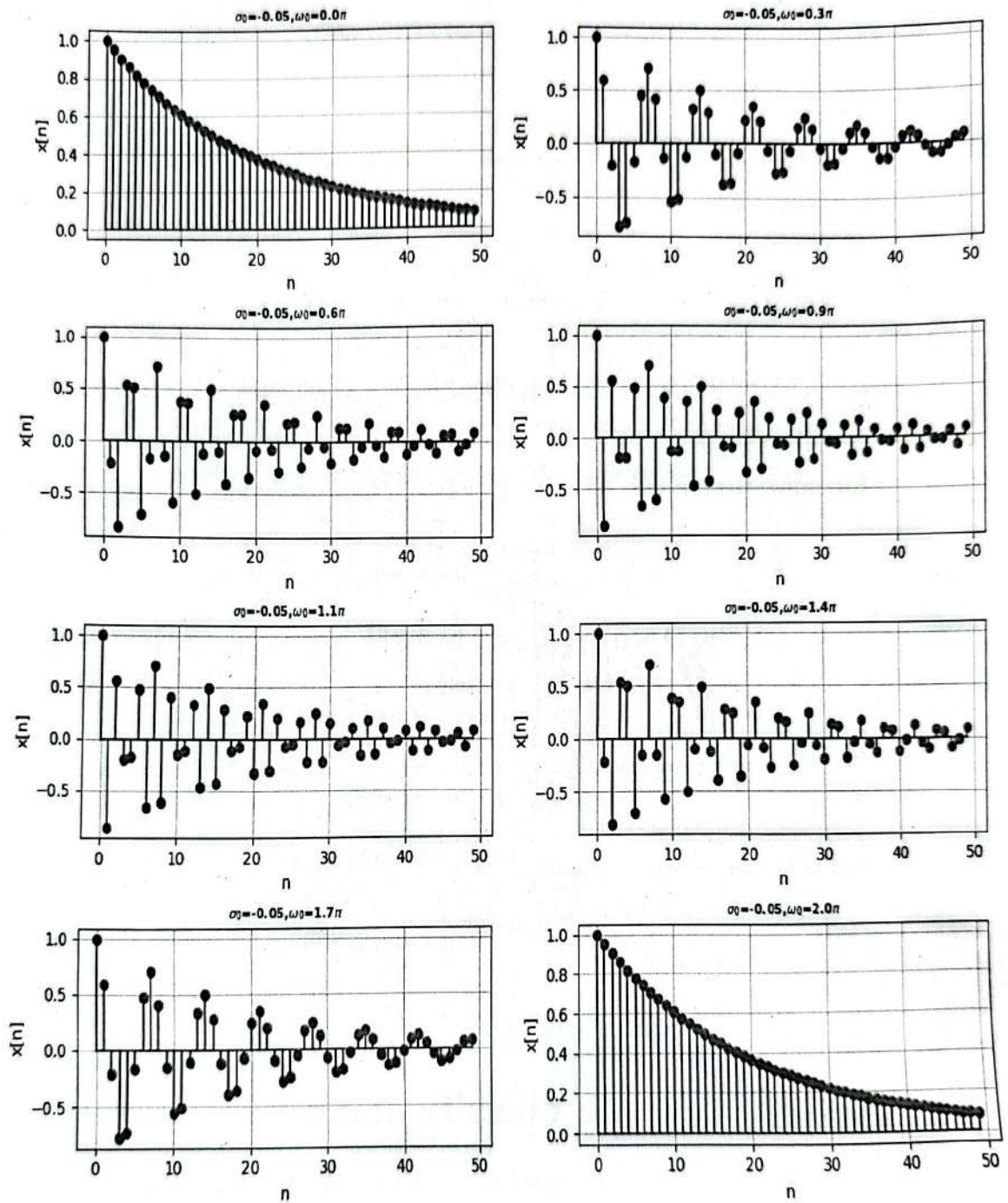


Figure 1.17: Plot of complex exponential sequence for various values of ω_0 , $\sigma_0 > 0$ and $n > 0$.

where,

a is scaling factor (integer for compression/expansion), and

b is integer shift.

Note: For the above transformation, time shifting must be performed before time scaling, i.e., first shift by b, giving gives $x[n + b]$ Then applying the scaling by a (substituting $n \rightarrow an$) yielding $x[an + b]$ directly, keeping all index operations integer.

Example: Given a discrete time signal as

$$x[n] = \begin{cases} 1, & \text{for } n = 0, \\ 2, & \text{for } n = 1, \\ 3, & \text{for } n = 2, \\ 4, & \text{for } n = 3, \\ 0, & \text{for elsewhere.} \end{cases} \quad (1.18)$$

Plot the sequence $y[n] = x[2n+1]$.

Answer: $y[n] = 2, 4$ for $n = 0, 1$.

Do for $y[n] = x[\frac{n}{2} + 2]$

1.4. Sequence classification

The classification of sequences can be carried out in different ways. One way to classify sequence is in terms of the **number of samples** defining the sequence. Based upon the number of samples in a given sequence, the sequence can be either finite length sequence or infinite length sequence. Example of finite

length and infinite length sequence are shown in Equation (1.19) and (1.20), respectively.

$$x = \{x[n]\} = \{3.4, 0.95, 2.15, 1.3, 2.94, -2.3, -3.5, 5.0, 7.4\} \quad (1.19)$$

↑

$$x = \{x[n]\} = \{\dots, 3.4, 0.95, 2.15, 1.3, 2.94, -2.3, -3.5, 5.0, 7.4, \dots\} \quad (1.20)$$

↑

Another classification of sequences is based upon the **symmetry** exhibited by the samples with respect to the time index $n = 0$. Furthermore, a discrete-time signal can also be classified based upon **periodicity**, **summability**, **energy**, and **power**.

1.4.1. Classification Based on Symmetry:

1.4.1.1. Even and Odd signal (sequence)

A real sequence is called an **even sequence** if it satisfies following criterion:

$$x[n] = x[-n]. \quad (1.21)$$

A real sequence is called an **odd sequence** if it satisfies following criterion:

$$x[n] = -x[-n] \implies -x[n] = x[-n]. \quad (1.22)$$

Any real sequence $x[n]$ can be expressed as a sum of its even part $x_{ev}[n]$ and its odd part $x_{od}[n]$:

$$x[n] = x_{ev}[n] + x_{od}[n] \quad (1.23)$$

Replacing n by $-n$ in equation (1.23), we get

$$x[-n] = x_{ev}[-n] + x_{od}[-n] \quad (1.24)$$

Using (1.21) and (1.22) in equation (1.24), we get

$$x[-n] = x_{ev}[n] - x_{od}[n] \quad (1.25)$$

Adding equation (1.23) and (1.25), we get

$$\begin{aligned} x[n] + x[-n] &= 2x_{ev}[n] + 0 \\ 2x_{ev}[n] &= x[n] + x[-n] \\ x_{ev}[n] &= \frac{1}{2}(x[n] + x[-n]). \end{aligned}$$

Similarly, subtracting equation (1.23) and (1.25), we get

$$\begin{aligned} x[n] - x[-n] &= 0 + 2x_{od}[n] \\ 2x_{od}[n] &= x[n] - x[-n] \\ x_{od}[n] &= \frac{1}{2}(x[n] - x[-n]). \end{aligned}$$

Decomposing a discrete time signal into its even and odd part:

A discrete time signal $x[n]$ can be expressed in terms of its even part ($x_e[n]$)

and odd part ($x_o[n]$), i.e.,

$$x_e[n] = \frac{1}{2}(x[n] + x[-n]) \quad (1.26)$$

$$x_o[n] = \frac{1}{2}(x[n] - x[-n]) \quad (1.27)$$

Note: In order to draw the even and odd part of the discrete time signal follow the steps:

- (i) Draw signal $x[n]$
- (ii) Draw the folded version of the signal, i.e. $x[-n]$
- (iii) Add $x[n]$ and $x[-n]$ and divide by 2 to get even part, i.e., $x_e[n]$ of $x[n]$.
- (iv) Subtract $x[-n]$ from $x[n]$ and divide by 2 to get odd part, i.e., $x_o[n]$ of $x[n]$.

Find and sketch even and odd part of the following discrete time sequence:

1. $x[n] = e^{-\frac{n}{4}}u[n]$

1. Solution: we can write $x[n]$ as:

$$x[n] = e^{-\frac{n}{4}}, n \geq 0 \text{ (since, } u[n] = 0 \text{ for } n < 0) \quad (1.28)$$

Step 1: Evaluating $x[n]$ for some values of n :

for $n = 0, 1, 2, 3$ we get,

$$x[0] = e^{-\frac{0}{4}} = 1$$

$$x[1] = e^{-\frac{1}{4}} = 0.78$$

$$x[2] = e^{-\frac{2}{4}} = 0.61$$

$$x[3] = e^{-\frac{3}{4}} = 0.47$$

$$x[4] = e^{-\frac{4}{4}} = 0.37$$

Step 2: Calculate $x[-n]$ from $x[n]$ we get,

$$x[-0] = x[0] = 1$$

$$x[-1] = x[1] = 0.78$$

$$x[-2] = x[2] = 0.61$$

$$x[-3] = x[3] = 0.47$$

$$x[-4] = x[4] = 0.37$$

$$x[n] = \{1, 0.78, 0.61, 0.47, 0.37\}$$

↑

$$x[-n] = \{0.37, 0.47, 0.61, 0.78, 1\}$$

↑

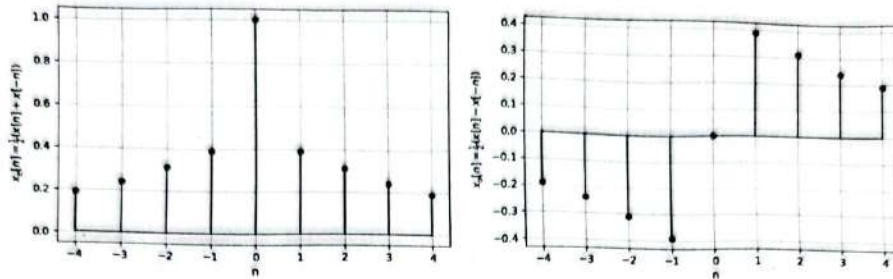


Figure 1.18: Sketch of the even and odd part of the given sequence $x[n]$

Step 3: calculate $x[n] + x[-n]$

$$x[n] + x[-n] = \{0.37, 0.47, 0.61, 0.78, 1 + 1, 0.78, 0.61, 0.47, 0.37\}$$

$$\begin{aligned} \text{We know, } x_e[n] &= \frac{1}{2}(x[n] + x[-n]) \\ &= \frac{1}{2}\{0.37, 0.47, 0.61, 0.78, 1 + 1, 0.78, 0.61, 0.47, 0.37\} \\ &= \{0.19, 0.24, 0.31, 0.39, 1, 0.39, 0.31, 0.24, 0.19\} \end{aligned}$$

Step 4: calculate $x[n] - x[-n]$

$$x[n] - x[-n] = \{-0.37, -0.47, -0.61, -0.78, 1 - 1, 0.78, 0.61, 0.47, 0.37\}$$

$$\begin{aligned} \text{We know, } x_o[n] &= \frac{1}{2}(x[n] - x[-n]) \\ &= \frac{1}{2}\{-0.37, -0.47, -0.61, -0.78, 1 - 1, 0.78, 0.61, 0.47, 0.37\} \\ &= \{-0.19, -0.24, -0.31, -0.39, 0, 0.39, 0.31, 0.24, 0.19\} \end{aligned}$$

Step 5: sketch signal $x_e[n]$ and $x_o[n]$. Please draw the signal yourself

roughly not to the scale by yourself

1.4.1.2. Conjugate-symmetric and Conjugate-antisymmetric sequence (signal)

A complex sequence $x[n]$ is called a **conjugate-symmetric** sequence if it satisfies the following criterion:

$$x[n] = x^*[-n]. \quad (1.29)$$

A sequence $x[n]$ is called a **conjugate-antisymmetric** sequence if it satisfies following criterion:

$$x[n] = -x^*[-n]. \quad (1.30)$$

Any complex sequence $x[n]$ can be expressed as a sum of its conjugate-symmetric part $x_{cs}[n]$ and its conjugate-antisymmetric part $x_{cas}[n]$:

$$x[n] = x_{cs}[n] + x_{cas}[n] \quad (1.31)$$

Replacing n by $-n$ in equation (1.31), we get

$$x[-n] = x_{cs}[-n] + x_{cas}[-n]$$

Taking conjugate

$$x^*[-n] = x_{cs}^*[-n] + x_{cas}^*[-n] \quad (1.32)$$

Using (1.29) and (1.30) in equation (1.32), we get

$$x^*[-n] = x_{cs}[n] - x_{cas}[n] \quad (1.33)$$

Adding equation (1.31) and (1.33), we get

$$\begin{aligned} x[n] + x^*[-n] &= 2x_{cs}[n] + 0, \\ 2x_{cs}[n] &= x[n] + x^*[-n], \\ x_{cs}[n] &= \frac{1}{2}(x[n] + x^*[-n]). \end{aligned}$$

Similarly, subtracting equation (1.31) and (1.33), we get

$$\begin{aligned} x[n] - x^*[-n] &= 0 + 2x_{cas}[n], \\ 2x_{cas}[n] &= x[n] - x^*[-n], \\ x_{cas}[n] &= \frac{1}{2}(x[n] - x^*[-n]). \end{aligned}$$

Example: Given a finite length sequence of length 7 defined for $-3 \leq n \leq 3$:

$$x[n] = \{0, 1 + 4j, -2 + 3j, 4 - 2j, -5 - 6j, -2j, 3\}$$

↑

Determine the conjugate-symmetric $x_{cs}[n]$ and the conjugate-antisymmetric part $x_{cas}[n]$ of $x[n]$.

Solution:

We know,

$$x_{cs}[n] = \frac{1}{2}(x[n] + x^*[-n])$$

and

$$x_{cas}[n] = \frac{1}{2}(x[n] - x^*[-n])$$

Step 1: Calculate $x^*[-n]$: We first calculate $x^*[n]$

$$x^*[n] = \{0, 1 - 4j, -2 - 3j, 4 + 2j, -5 + 6j, 2j, 3\}$$

↑

After time folding/reversal of $x^*[n]$, we get

$$x^*[-n] = \{3, 2j, -5 + 6j, 4 + 2j, -2 - 3j, 1 - 4j, 0\}$$

↑

Using first equation we get,

$$x_{cs}[n] = \frac{1}{2}\{0 + 3, 1 + 4j + 2j, -2 + 3j - 5 + 6j, 4 - 2j + 4 + 2j, \\ -5 - 6j - 2 - 3j, -2j + 1 - 4j, 3 + 0\}$$

$$x_{cs}[n] = \{1.5, 0.5 + 3j, -3.5 + 4.5j, 4, -3.5 - 4.5j, 0.5 - 3j, 1.5\}$$

↑

Similarly, after some algebraic manipulation we get,

$$x_{cas}[n] = \{-1.5, 0.5 + j, 1.5 - 1.5j, -2j, -1.5 - 1.5j, -0.5 + j, 1.5\}$$

↑

1.4.2. Classification Based on Periodicity

Periodic sequence: A sequence $x[n]$ is said to be periodic if it satisfies the following condition:

$$x[n] = x[n + kN], \quad (1.34)$$

where,

$k \in \mathbb{Z}$, and

$N \in \mathbb{Z}^+$.

Aperiodic/non-periodic: A sequence is aperiodic or non-periodic if it is not periodic sequence, i.e., aperiodic signal does not satisfy equation (1.35). The smallest value of N for which equation (1.35) holds true is called fundamental period denoted as N_f .

Condition for Periodicity of discrete time signal: A discrete time signal is periodic if the frequency f_0 is a rational number, i.e., the frequency f_0 should be in the form of ratio of two integers:

From the definition of periodicity, $x[n]$ is periodic if:

$$x[n] = x[n + N] \quad (1.35)$$

Let $x[n]$ be a sinusoidal signal given as:

$$x[n] = A \cos(\omega_0 n + \phi) \quad (1.36)$$

From equation (1.35) and (1.36)

$$x[n + N] = A \cos(\omega_o(n + N) + \phi) \quad (1.37)$$

$$= A \cos(\omega_o n + \omega_o N + \phi) \quad (1.38)$$

For $x[n + N]$ to be periodic:

$$\omega_o N = 2\pi K \implies N = \frac{2\pi K}{\omega_o} \implies f_o = \frac{K}{N} \quad (1.39)$$

Periodicity condition for the discrete time signal of the form $x[n] = x_1[n] + x_2[n]$ or $x_1[n].x_2[n]$: If the given input sequence is expressed in terms of either summation or multiplication of two different sequence. We first calculate the values of f_1 and f_2 corresponding to the sequences $x_1[n]$ and $x_2[n]$, respectively. Two case arises:

- x_1 and x_2 are both periodic: If both x_1 and x_2 are periodic then we get:

$$f_1 = \frac{k_1}{N_1}$$

and

$$f_2 = \frac{k_2}{N_2}$$

The fundamental period of $x[n] = LCM(N_1, N_2)$

- Either x_1 or x_2 are non-periodic: Then the signal is non-periodic.

Periodicity of Discrete time signal: A discrete time sinusoidal signal is periodic only if its frequency ω_0 is rational. This means that frequency ω_0 should be in the form of ratio of two integers.

– **Periodicity Condition for $x[n]$:**

The input sequence is a single discrete time sequences.

1. Step 1: Find f_0 . Comparing $x[n]$ with either $\cos(\omega_0 n)$ or $\sin(\omega_0 n)$.

$$\omega_0 = \dots$$

$$f_0 = \frac{\dots}{2\pi}$$

2. Step 2: If f_0 is rational, i.e., ratio of two integer. $x[n]$ is periodic.

* Fundamental period (N):

$$f_0 = \frac{k}{N} \quad (1.40)$$

The fundamental period is the value of N.

3. If f_0 is irrational. $x[n]$ is non-periodic.

– **Periodicity Condition for $x[n] = x_1[n] + x_2[n]$:**

The input sequence is expressed as summation of two discrete time sequences.

1. Calculate values of f_1 for $x_1[n]$, f_2 for $x_2[n]$.

(a) If both $x_1[n]$ and $x_2[n]$ are periodic find fundamental period N_1 and N_2 corresponding to sequence $x_1[n]$ and $x_2[n]$

i. If: $\frac{N_1}{N_2}$ is ratio of two integers:

$x[n]$ is periodic fundamental period (N) is LCM (N_1, N_2)

ii. Else: Not periodic

1. Determine whether the signal $x[n] = \cos[\pi n/2].\cos[\pi n/4]$ is periodic or non-periodic, and if it is periodic, find its fundamental period. (2078

Bhadra)

Solution:

Given

$$x[n] = \cos[\pi n/2].\cos[\pi n/4]$$

$$= \frac{1}{2}\{2\cos[\pi n/2].\cos[\pi n/4]\}$$

$$= \frac{1}{2}\{\cos(\frac{\pi n}{2} + \frac{\pi n}{4}) + \cos(\frac{\pi n}{2} - \frac{\pi n}{4})\}$$

$$= \frac{1}{2}\{\cos(\frac{3\pi n}{4}) + \cos(\frac{\pi n}{4})\}$$

Let $x_1[n] = \frac{1}{2}\cos(\frac{3\pi n}{4})$ and $x_2[n] = \frac{1}{2}\cos(\frac{\pi n}{4})$.

Comparing these equations with $x[n] = A\cos(\omega_0 n + \phi)$

Here,

$$\omega_1 = \frac{3\pi}{4}$$

and

$$\omega_2 = \frac{\pi}{4}$$

So,

$$\frac{2\pi}{\omega_1} = \frac{2\pi}{\frac{3\pi}{4}} = \frac{8}{3}$$

and

$$\frac{2\pi}{\omega_2} = \frac{2\pi}{\frac{\pi}{4}} = 8$$

Since both the results are rational then both $x_1[n]$ and $x_2[n]$ signals are periodic meaning that $x[n]$ is periodic.

Now to calculate the fundamental period, we have

For $x_1[n]$

$$\frac{2\pi}{\omega_1} = \frac{N_1}{k} = \frac{8}{3}$$

For N_1 to be an integer we put $k = 3$ and we get

$$N_1 = \frac{8}{3} * 3 = 8$$

Similarly, for $x_2[n]$

$$\frac{2\pi}{\omega_1} = \frac{N_2}{k} = 8$$

$$\therefore N_2 = 8$$

Therefore, the fundamental period for the signal $x[n]$ is

$$N = LCM(N_1, N_2) = LCM(8, 8) = 8$$

3. Define energy and power signal. Determine whether the signal $x[n] = \cos[\frac{2\pi n}{5}] + \sin[\frac{\pi n}{3}]$ is periodic or non-periodic and if it is periodic, find its fundamental period. (2079 Bhadra, 2081 Bhadra)

Solution:

Let $x[n]$ be a type of discrete time signal. Then the signal $x[n]$ is said to be energy signal if its total energy is finite and average power is zero.

Similarly the signal $x[n]$ is said to be power signal if its total average power is finite and total energy is infinite.

Total Energy of a signal is given by:

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

For periodic signal $x[n]$, the average power is given by:

$$P_{avg} = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2$$

Given

$$x[n] = \cos[\frac{2\pi n}{5}] + \sin[\frac{\pi n}{3}]$$

Comparing with $x[n] = A_1 \cos(\omega_1 n + \phi_1) + A_2 \sin(\omega_2 n + \phi_2)$

We get,

$$\omega_1 = \frac{2\pi}{5}, \omega_2 = \frac{\pi}{3}$$
$$\therefore \frac{2\pi}{\omega_1} = 5, \frac{2\pi}{\omega_2} = 6$$

Since both the results are rational, the signal $x_1[n]$ and $x_2[n]$ are periodic.

This means the signal $x[n]$ is also periodic.

Now to calculate fundamental period we have,

$$\frac{2\pi}{\omega_1} = \frac{N_1}{k} = 5$$

$$\therefore N_1 = 5$$

Similarly,

$$\frac{2\pi}{\omega_2} = \frac{N_2}{k} = 6$$

$$\therefore N_2 = 6$$

Thus, the fundamental period for the signal $x[n]$ is

$$N = LCM(N_1, N_2) = LCM(5, 6) = 30$$

6. Determine whether the signal is periodic or not. If the signal is peri-

odic, determine the fundamental period $x[n] = e^{\frac{j\pi n}{16}} \cos\{\frac{n\pi}{17}\}$. (2080 Bhadra)

Solution:

Given

$$x[n] = e^{\frac{j\pi n}{16}} \cos\{\frac{n\pi}{17}\}$$

Using Euler's identity

$$x[n] = [\cos(\frac{\pi n}{16}) + j\sin(\frac{\pi n}{16})]\cos(\frac{\pi n}{17})$$

Disregarding the imaginary part as it is a scaling factor and not a signal.

$$\begin{aligned} x[n] &= \cos(\frac{\pi n}{16})\cos(\frac{\pi n}{17}) \\ &= \frac{1}{2}[2\cos(\frac{\pi n}{16})\cos(\frac{\pi n}{17})] \\ &= \frac{1}{2}[\cos(\frac{\pi n}{16} + \frac{\pi n}{17}) + \cos(\frac{\pi n}{16} - \frac{\pi n}{17})] \\ &= \frac{1}{2}[\cos(\frac{33\pi n}{272}) + \cos(\frac{\pi n}{272})] \end{aligned}$$

Let $x_1[n] = \frac{1}{2}\cos(\frac{33\pi n}{272})$ and $x_2[n] = \frac{1}{2}\cos(\frac{\pi n}{272})$.

Comparing with $x[n] = A\cos(w_0n + \phi)$ we get,

$$\begin{aligned} w_1 &= \frac{33\pi}{272}, w_2 = \frac{\pi}{272} \\ \therefore \frac{2\pi}{\omega_1} &= \frac{544}{33}, \frac{2\pi}{\omega_2} = 544 \end{aligned}$$

Since both the results are rational, the signal $x_1[n]$ and $x_2[n]$ are periodic.

This means the signal $x[n]$ is also periodic.

Now to calculate fundamental period we have,

$$\frac{2\pi}{\omega_1} = \frac{N_1}{k} = \frac{544}{33}$$

$$\therefore N_1 = 544$$

Similarly,

$$\frac{2\pi}{\omega_2} = \frac{N_2}{k} = 544$$

$$\therefore N_2 = 544$$

Thus, the fundamental period for the signal $x[n]$ is

$$N = LCM(N_1, N_2) = LCM(544, 544) = 544$$

Determine whether the following discrete-time signals are periodic or not? If periodic determine fundamental period.

1. $\cos(3\pi n)$
2. $\sin(3n)$
3. $\cos(\frac{2\pi n}{5}) + \cos(\frac{2\pi n}{7})$

4. $\cos(\frac{n}{8}) \cos(\frac{\pi n}{8})$

5. $\sin(\pi + 0.2n)$

6. $e^{j\frac{\pi}{4}n}$

7. $\cos(\frac{\pi n^2}{8})$

8. $R_e\{e^{j\frac{n\pi}{12}}\} + I_m\{e^{j\frac{n\pi}{18}}\}$ Ans: (N1=24, N2=36, N=12)

9. $e^{j\frac{\pi}{16}n} \cos\{\frac{n\pi}{17}\}$ Ans: (N1=32, N2=34, N = LCM(N1,N2))

– **Solution 1:** Given discrete time sequence is $x[n] = \cos(3\pi n)$

Comparing above expression with $x[n] = \cos(\omega_0 n)$, we have

$$\omega_0 n = 3\pi n$$

$$2\pi f_0 = 3\pi \text{ (since, } \omega_0 = 2\pi f_0 \text{)}$$

$$f_0 = \frac{3}{2}$$

Since $f_0 = \frac{3}{2}$, is rational or ratio of two integer. The given signal is periodic.

Fundamental period (N): Here, $f_0 = \frac{k}{N} = \frac{3}{2}$. The fundamental period is 2.

– **Solution 2:** Given discrete time sequence is $x[n] = \sin(3n)$

Comparing above expression with $x[n] = \sin(\omega_0 n)$, we have

$$\omega_0 n = 3n$$

$$2\pi f_0 = 3 \text{ (since, } \omega_0 = 2\pi f_0 \text{)}$$

$$f_0 = \frac{3}{2\pi}$$

Since $f_0 = \frac{3}{2\pi}$, is irrational or not ratio of two integer. The given signal is non-periodic or aperiodic.

– **Solution 3:** Given discrete time sequence is $x[n] = \cos(\frac{2\pi n}{5}) + \cos(\frac{2\pi n}{7})$

Comparing above expression with $x[n] = \cos(\omega_1 n) + \cos(\omega_2 n)$, we have

$$\omega_1 n = 3n$$

$$2\pi f_1 = 3 \text{ (since, } \omega_1 = 2\pi f_1 \text{)}$$

$$f_1 = \frac{1}{5}$$

and

$$\omega_2 n = 3n$$

$$2\pi f_2 = 3 \text{ (since, } \omega_2 = 2\pi f_2 \text{)}$$

$$f_2 = \frac{1}{7}$$

$f_1 = \frac{1}{5}$ is rational, therefore, $\cos(\frac{2\pi n}{5})$ is periodic and

$f_2 = \frac{1}{7}$ is rational, $\cos(\frac{2\pi n}{7})$ is also periodic.

Fundamental period of $\cos(\frac{2\pi n}{5})(N_1)$: Here $f_1 = \frac{k}{N_1} = \frac{1}{5}$

Fundamental period of $\cos(\frac{2\pi n}{7})(N_2)$: Here $f_2 = \frac{k}{N_2} = \frac{1}{7}$

The fundamental period of $\cos(\frac{2\pi n}{5})$ is 5.

The fundamental period of $\cos(\frac{2\pi n}{7})$ is 7.

Since, $\frac{N_1}{N_2} = \frac{5}{7}$ is rational the given sequence is periodic.

The fundamental period of $x[n] = \text{LCM}(N_1, N_2) = \text{LCM}(5, 7) = 35$

– **Solution 4:** Given discrete time sequence is $\cos(\frac{n}{8}) + \cos(\frac{\pi n}{8})$

Comparing above expression with $x[n] = \cos(\omega_1 n) + \cos(\omega_2 n)$, we have

$$\omega_1 n = \frac{n}{8}$$

$$2\pi f_1 = 3 \text{ (since, } \omega_1 = 2\pi f_1 \text{)}$$

$$f_1 = \frac{1}{16\pi}$$

and

$$\omega_2 n = \frac{\pi n}{8}$$

$$2\pi f_2 = 3 \text{ (since, } \omega_2 = 2\pi f_2 \text{)}$$

$$f_2 = \frac{1}{16}$$

$f_1 = \frac{1}{16\pi}$ is irrational, $\cos(\frac{n}{8})$ is non-periodic and,

$f_2 = \frac{1}{16}$ is rational, $\cos(\frac{\pi n}{8})$ is periodic.

Here, $\cos(\frac{n}{8})$ is periodic and $\cos(\frac{\pi n}{8})$ is non-periodic. Therefore, $x[n]$ is non-periodic since, it is the product of periodic and non-periodic sequence.

– **Solution 5:** Given discrete time sequence is $\sin(\pi + 0.2n)$

Comparing above expression with $x[n] = \sin(\omega_1 n + \phi)$ we have

$$\omega_0 n = 0.2n$$

$$2\pi f_0 = 0.2 \text{ (since, } \omega_0 = 2\pi f_0 \text{)}$$

$$f_0 = \frac{1}{10\pi}$$

$f_0 = \frac{1}{10\pi}$ is irrational, therefore, $x[n]$ is non-periodic.

– **Solution 6:** Given discrete time sequence is $e^{j\frac{\pi}{4}n}$

Using Euler's identity:

$$x[n] = \cos\left(\frac{\pi}{4}n\right) + j \sin\left(\frac{\pi}{4}n\right)$$

Comparing above expression with $x[n] = \cos(\omega_0 n) + j \sin(\omega_0 n)$ we have

$$\begin{aligned}\omega_0 n &= \frac{\pi}{4}n \\ 2\pi f_0 &= \frac{\pi}{4} \text{ (since, } \omega_0 = 2\pi f_0 \text{)} \\ f_0 &= \frac{1}{8}\end{aligned}$$

$f_0 = \frac{1}{8}$ is rational, therefore, $x[n]$ is periodic.

– **Solution 7:** Given discrete time sequence is $\cos(\frac{\pi n^2}{8})$

We cannot do this as previous solution because we have n^2 term inside bracket. Therefore, we go by definition, for $x[n]$ to be periodic

$$x[n] = x[n + N] \quad (1.41)$$

$$\begin{aligned}x[n + N] &= \cos\left(\frac{\pi}{8}(n + N)^2\right) \\ &= \cos\left(\frac{\pi}{8}(n^2 + 2nN + N^2)\right) \\ &= \cos\left(\frac{\pi}{8}n^2 + \frac{\pi}{8}2nN + \frac{\pi}{8}N^2\right)\end{aligned}$$

For periodicity, i.e. $x[n] = x[n + N]$

$$\cos\left(\frac{\pi n^2}{8}\right) = \cos\left(\frac{\pi}{8}n^2 + \frac{\pi}{8}2nN + \frac{\pi}{8}N^2\right) \quad (1.42)$$

For Equation 1.42 to be true,

* $\frac{\pi}{8}N^2 = 2\pi m$, for N to be integer $\Rightarrow m=4$ which gives $N=8$ (Note: if you put any other smaller values of m less than 4 you dont get N as integer.)

* $\frac{\pi}{8}2N = 2\pi k$ for N to be integer $\Rightarrow k=1$

Hence, the given signal is periodic with period $N=8$.

1.4.3. Classification based upon energy and power of sequence:

1.4.3.1. Energy sequence (signal)

Let $x[n]$ be a sequence, then the energy of the sequence is defined as:

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2. \quad (1.43)$$

1.4.3.2. Power sequence (signal)

The average power of a periodic sequence with period N is given as:

$$P_{avg} = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2. \quad (1.44)$$

The average power of an aperiodic sequence is given as:

$$P = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K |x[n]|^2. \quad (1.45)$$

Note: A power signal has finite average power and infinite energy.

An energy signal has finite energy and zero power. 7. Compare between energy signal and power signal. Determine whether the signal $x[n] = e^{j(\frac{n^2}{4} + \frac{\pi n}{4})}$ is an energy signal or a power signal. [2082 Bhadra]

Solution:

Part 1: Comparison between Energy Signal and Power Signal

Aspect	Energy Signal	Power Signal
Definition	An energy signal is a signal that has finite energy and zero average power over an infinite time interval. Mathematically: $E = \sum_{n=-\infty}^{\infty} x[n] ^2 < \infty$	A power signal is a signal that has finite average power and infinite energy over an infinite time interval. Mathematically: $P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x[n] ^2$ and $P = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} x[n] ^2$
Energy and Power	Energy signals have finite energy ($0 < E < \infty$) and zero average power ($P = 0$).	Power signals have infinite energy ($E = \infty$) and finite average power ($0 < P < \infty$).
Periodicity	Energy signals are typically non-periodic, transient, or have finite duration. They decay to zero as time approaches infinity.	Power signals are typically periodic signals or signals with constant amplitude that do not decay to zero.
Example	Impulse function: $x[n] = \delta[n]$; Finite duration signal: $x[n] = u[n] - u[n-5]$	Sinusoidal signal: $x[n] = \cos(\omega_0 n)$; Constant signal: $x[n] = A$

Part 2: Analysis of the given signal

Given: $x[n] = e^{j(\frac{n^2}{4} + \frac{\pi n}{4})}$

First, let's find the magnitude:

$$|x[n]| = \left| e^{j(\frac{n^2}{4} + \frac{\pi n}{4})} \right| = 1$$

Since the magnitude is constant and equal to 1 for all n :

Energy calculation:

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \sum_{n=-\infty}^{\infty} 1^2 = \infty$$

Power calculation:

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2 = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N 1 = \lim_{N \rightarrow \infty} \frac{2N+1}{2N+1} = 1$$

Since $E = \infty$ and $P = 1$ (finite), the signal $x[n] = e^{j(\frac{n^2}{4} + \frac{\pi n}{4})}$ is a **power signal**.

1.5. Discrete time System (DTS)

A Discrete time system processes an input sequence $x[n]$ and produces an output sequence $y[n]$. A discrete time system is defined in terms of relation between input and output sequence, which is usually specified by a formula. Mathematically, a discrete time system is defined as a transformation of input sequence $x[n]$ to output sequence $y[n]$, i.e.,

$$y[n] = T\{x[n]\} \quad (1.46)$$

Pictorially, a DTS is shown in Figure 1.19. All the basic operation on discrete time signal can be considered to be carried out by appropriate devices. Such devices are called elementary discrete time system.

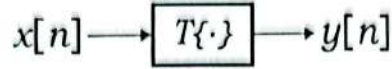


Figure 1.19: A Discrete Time System.

1.5.1. Classification of Discrete time system

Depending upon the input and output relation of a discrete time system, it can be classified in a number of ways:

1. **Static and dynamic discrete time system:** A discrete time system is said to be static if the output at any time instant depends on input samples at the same time. Examples:

$$y[n] = 20x[n]$$

$$y[n] = 20x^2[n] + 15x[n] + 100$$

A discrete time system is said to be dynamic if the output at any time instant depends on input sample at the same time instant as well as other time instant. Examples:

$$y[n] = 10x[n] + 20x[n - 1]$$

$$y[n] = 3x[n + 2] + x[n]$$

2. **Linear and non-linear discrete time system:** A discrete time system is said to be linear if it holds superposition principle. Let $y_1[n] = T\{x_1[n]\}$ and $y_2[n] = T\{x_2[n]\}$ be the output or response of system (T) for inputs

$x_1[n]$ and $x_2[n]$, separately, i.e.,

$$y_1[n] = T\{x_1[n]\}$$

$$y_2[n] = T\{x_2[n]\}$$

The system T is linear if and only if:

$$T\{x_1[n] + x_2[n]\} = T\{x_1[n]\} + T\{x_2[n]\}$$

$$T\{ax[n]\} = aT\{x[n]\}$$

or,

$$T\{ax_1[n] + bx_2[n]\} = aT\{x_1[n]\} + bT\{x_2[n]\} \quad (1.47)$$

Determine whether following system is linear or non-linear:

In order to solve this question, follow the following step:

- Step 1: Calculate $a_1\mathcal{T}\{x_1[n]\} + a_2\mathcal{T}\{x_2[n]\}$

$$\mathcal{T}\{x_1[n]\} = \dots$$

$$\mathcal{T}\{x_2[n]\} = \dots$$

$$a_1\mathcal{T}\{x_1[n]\} + a_2\mathcal{T}\{x_2[n]\} = \dots$$

- Step 2: Calculate $\mathcal{T}\{a_1x_1[n] + a_2x_2[n]\}$

$$\mathcal{T}\{a_1x_1[n] + a_2x_2[n]\} = \dots$$

- Step 3:

– If $a_1\mathcal{T}\{x_1[n]\} + a_2\mathcal{T}\{x_2[n]\} = \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\}$ the system is linear.

– Else the system is non-linear.

(a) $Ax[n] + B$

(b) $y[n] = nx[n]$

- (c) $y[n] = e^{x[n]}$
- (d) $y[n] = x^2[n]$
- (e) $y[n] = x[n^2]$
- (f) $y[n] = x[-n]$
- (g) $y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n+3])$
- (h) $y[n] = x[n] \cos(\omega_0 n)$
- (i) $y[n] = \cos(\frac{2\pi}{8}n + \frac{\pi}{4})$

• **Solution 1:** Given System is $y[n] = Ax[n] + B$

Step 1: Calculate output of system for two inputs $x_1[n]$ and $x_2[n]$

$$\mathcal{T}\{x_1[n]\} = Ax_1[n] + B \text{ and}$$

$$\mathcal{T}\{x_2[n]\} = Ax_2[n] + B \text{ Hence,}$$

$$\mathcal{T}\{a_1x_1[n]\} + \mathcal{T}\{a_2x_2[n]\} = a_1Ax_1[n] + B + a_2Ax_2[n] + B$$

Step 2: Calculate output of the system for linear combination of two inputs $x_1[n]$ and $x_2[n]$, i.e., $a_1x_1[n] + a_2x_2[n]$

$$\begin{aligned} \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\} &= A\{a_1x_1[n] + a_2x_2[n]\} + B \\ &= Aa_1x_1[n] + Aa_2x_2[n] + B \end{aligned}$$

Since, $\mathcal{T}\{x_1[n]\} + \mathcal{T}\{x_2[n]\} \neq \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\}$. Hence the system is non-linear.

• **Solution 2:** Given System is $y[n] = nx[n]$

Step 1: Calculate output of system for two inputs $x_1[n]$ and $x_2[n]$

$$\mathcal{T}\{x_1[n]\} = nx_1[n]$$

$$\mathcal{T}\{x_2[n]\} = nx_2[n]$$

$$\text{Therefore, } \mathcal{T}\{a_1x_1[n]\} + \mathcal{T}\{a_2x_2[n]\} = a_1nx_1[n] + a_2nx_2[n]$$

Step 2: Calculate output of the system for linear combination of two inputs $x_1[n]$ and $x_2[n]$, i.e., $a_1x_1[n] + a_2x_2[n]$

$$\begin{aligned}\mathcal{T}\{a_1x_1[n] + a_2x_2[n]\} &= n\{a_1x_1[n] + a_2x_2[n]\} \\ &= na_1x_1[n] + na_2x_2[n]\end{aligned}$$

Since, $\mathcal{T}\{x_1[n]\} + \mathcal{T}\{x_2[n]\} = \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\}$. Hence the system is linear.

- **Solution 3:** Given System is $y[n] = e^{x[n]}$

Step 1: Calculate output of system for two inputs $x_1[n]$ and $x_2[n]$

$$\mathcal{T}\{x_1[n]\} = e^{x_1[n]}$$

$$\mathcal{T}\{x_2[n]\} = e^{x_2[n]}$$

$$\text{Therefore, } \mathcal{T}\{a_1x_1[n]\} + \mathcal{T}\{a_2x_2[n]\} = e^{a_1x_1[n]} + e^{a_2x_2[n]}$$

Step 2: Calculate output of the system for linear combination of two inputs $x_1[n]$ and $x_2[n]$, i.e., $a_1x_1[n] + a_2x_2[n]$

$$\begin{aligned}\mathcal{T}\{a_1x_1[n] + a_2x_2[n]\} &= e^{(a_1x_1[n] + a_2x_2[n])} \\ &= e^{a_1x_1[n]}e^{a_2x_2[n]}\end{aligned}$$

Since, $\mathcal{T}\{x_1[n]\} + \mathcal{T}\{x_2[n]\} \neq \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\}$. Hence, the system is non-linear.

- **Solution 4:** Given System is $y[n] = x^2[n]$

Step 1: Calculate output of system for two inputs $x_1[n]$ and $x_2[n]$

$$\mathcal{T}\{x_1[n]\} = (x_1[n])^2 = x_1^2[n]$$

$$\mathcal{T}\{x_2[n]\} = (x_2[n])^2 = x_2^2[n]$$

$$\text{Therefore, } \mathcal{T}\{a_1x_1[n]\} + \mathcal{T}\{a_2x_2[n]\} = a_1^2x_1^2[n] + a_2^2x_2^2[n]$$

Step 2: Calculate output of the system for linear combination of two inputs $x_1[n]$ and $x_2[n]$, i.e., $a_1x_1[n] + a_2x_2[n]$

$$\begin{aligned} \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\} &= (a_1x_1[n] + a_2x_2[n])^2 \\ &= a_1^2x_1^2[n] + 2a_1x_1[n]a_2x_2[n] + a_2^2x_2^2[n] \end{aligned}$$

Since, $\mathcal{T}\{x_1[n]\} + \mathcal{T}\{x_2[n]\} \neq \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\}$. Hence, the system is non-linear.

- **Solution 5:** Given System is $y[n] = x[n^2]$

Step 1: Calculate output of system for two inputs $x_1[n]$ and $x_2[n]$

$$\mathcal{T}\{x_1[n]\} = (x_1[n^2])$$

$$\mathcal{T}\{x_2[n]\} = (x_2[n^2])$$

$$\text{Therefore, } \mathcal{T}\{a_1x_1[n]\} + \mathcal{T}\{a_2x_2[n]\} = a_1x_1[n^2] + a_2x_2[n^2]$$

Step 2: Calculate output of the system for linear combination of two

inputs $x_1[n]$ and $x_2[n]$, i.e., $a_1x_1[n] + a_2x_2[n]$

$$\begin{aligned}\mathcal{T}\{a_1x_1[n] + a_2x_2[n]\} &= (a_1x_1[n] + a_2x_2[n])^2 \\ &= a_1x_1[n^2] + a_2x_2[n^2]\end{aligned}$$

Since, $\mathcal{T}\{x_1[n]\} + \mathcal{T}\{x_2[n]\} = \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\}$. Hence the system is linear.

- **Solution 6:** Given System is $y[n] = x[-n]$

Step 1: Calculate output of system for two inputs $x_1[n]$ and $x_2[n]$

$$\mathcal{T}\{x_1[n]\} = (x_1[-n])$$

$$\mathcal{T}\{x_2[n]\} = (x_2[-n])$$

$$\text{Therefore, } \mathcal{T}\{a_1x_1[n]\} + \mathcal{T}\{a_2x_2[n]\} = a_1x_1[-n] + a_2x_2[-n]$$

Step 2: Calculate output of the system for linear combination of two inputs $x_1[n]$ and $x_2[n]$, i.e., $a_1x_1[n] + a_2x_2[n]$

$$\mathcal{T}\{a_1x_1[n] + a_2x_2[n]\} = a_1x_1[-n] + a_2x_2[-n]$$

Since, $\mathcal{T}\{x_1[n]\} + \mathcal{T}\{x_2[n]\} = \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\}$. Hence, the system is linear.

- **Solution 7:** Given System is $\mathcal{T}\{x[n]\} = \frac{1}{3}(x[n] + x[n-1] + x[n+3])$

Step 1: Calculate output of system for two inputs $x_1[n]$ and $x_2[n]$

$$\mathcal{T}\{x_1[n]\} = \frac{1}{3}(x_1[n] + x_1[n-1] + x_1[n+3])$$

$$\mathcal{T}\{x_2[n]\} = \frac{1}{3}(x_2[n] + x_2[n-1] + x_2[n+3])$$

$$\begin{aligned} \text{Therefore, } \mathcal{T}\{a_1x_1[n]\} + \mathcal{T}\{a_2x_2[n]\} &= \frac{1}{3}(a_1x_1[n] + a_1x_1[n-1] + a_1x_1[n+3]) + \\ &\quad \frac{1}{3}(a_2x_2[n] + a_2x_2[n-1] + a_2x_2[n+3]) \end{aligned}$$

Step 2: Calculate output of the system for linear combination of two inputs $x_1[n]$ and $x_2[n]$, i.e., $a_1x_1[n] + a_2x_2[n]$

$$\begin{aligned} \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\} &= \frac{1}{3}(a_1x_1[n] + a_2x_2[n] + a_1x_1[n-1] + \\ &\quad a_2x_2[n-1] + a_1x_1[n+3] + a_2x_2[n+3]) \\ &= \frac{1}{3}(a_1x_1[n] + a_1x_1[n-1] + a_1x_1[n+3]) + \\ &\quad \frac{1}{3}(a_2x_2[n] + a_2x_2[n-1] + a_2x_2[n+3]) \end{aligned}$$

Since, $\mathcal{T}\{x_1[n]\} + \mathcal{T}\{x_2[n]\} = \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\}$. Hence, the system is linear.

- **Solution 8:** Given System is $\mathcal{T}\{x[n]\} = x[n] \cos(\omega_0 n)$

Step 1: Calculate output of system for two inputs $x_1[n]$ and $x_2[n]$

$$\mathcal{T}\{x_1[n]\} = x_1[n] \cos(\omega_0 n)$$

$$\mathcal{T}\{x_2[n]\} = x_2[n] \cos(\omega_0 n)$$

$$\text{Therefore, } \mathcal{T}\{a_1x_1[n]\} + \mathcal{T}\{a_2x_2[n]\} = a_1x_1[n] \cos(\omega_0 n) + a_2x_2[n] \cos(\omega_0 n)$$

Step 2: Calculate output of the system for linear combination of two

inputs $x_1[n]$ and $x_2[n]$, i.e., $a_1x_1[n] + a_2x_2[n]$

$$\begin{aligned}\mathcal{T}\{a_1x_1[n] + a_2x_2[n]\} &= (a_1x_1[n] + a_2x_2[n]) \cos(\omega_0n) \\ &= a_1x_1[n] \cos(\omega_0n) + a_2x_2[n] \cos(\omega_0n)\end{aligned}$$

Since, $\mathcal{T}\{x_1[n]\} + \mathcal{T}\{x_2[n]\} = \mathcal{T}\{a_1x_1[n] + a_2x_2[n]\}$. Hence, the system is linear.

3. **Time-invariant** and time-variant system: Let $y[n]$ be the output of a discrete-time system for input $x[n]$, then the system is said to be time-invariant system if input $x[n - k]$, i.e., delayed by k units, gives output $y[n - k]$, i.e., output is also delayed by the same k units. For time-invariant system

$$\begin{aligned}\text{if } y[n] &= x[n], \text{ then,} \\ y[n - k] &= x[n - k].\end{aligned}$$

Determine whether following system is time invariant or time variant:

In order to solve this question, follow the following step:

- Step 1: First delay input $x[n]$ by k units, i.e. replace all occurrence of $x[n]$ by $x[n - k]$. Represent this equation as $y[n, k]$. Note $x[-n]$ will be $x[-n - k]$, this is because we want whole input sequence by k units, (Do not replace all occurrence of n by $n - k$).
- Step 2: Replace all occurrence of n by $n - k$. Represent this equation as $y[n - k]$.
- Step 3: if $y[n, k] = y[n - k]$ the system is time invariant else if $y[n, k] \neq$

$y[n - k]$ the system is time variant.

(a) $y[n] = x[n] - x[n - 1]$

(b) $y[n] = x[n] - nx[n - 1]$

(c) $y[n] = \cos(x[n])$

(d) $y[n] = |x[n]|$

(e) $y[n] = x[-n]$

(f) $y[n] = y[n - 4] + x[n - 4]$

- **Solution 1:** Given discrete time system is $y[n] = x[n] - x[n - 1]$

Step 1: Delay input by k units, we get

$$y[n, k] = x[n - k] - x[n - k - 1] \quad (1.48)$$

Step 2: Replace all n by $n - k$, we get

$$y[n - k] = x[n - k] - x[n - k - 1] \quad (1.49)$$

Since, $y[n, k] = y[n - k]$, the given system time invariant.

- **Solution 2:** Given discrete time system is $y[n] = x[n] - nx[n - 1]$

Step 1: Delay input by k units, we get

$$y[n, k] = x[n - k] - nx[n - k - 1] \quad (1.50)$$

Step 2: Replace all n by $n - k$, we get

$$\begin{aligned} y[n - k] &= x[n - k] - (n - k)x[n - k - 1] \\ &= x[n - k] - nx[n - k - 1] + kx[n - k - 1] \end{aligned}$$

Since, $y[n, k] \neq y[n - k]$, the given system time variant.

- **Solution 3:** Given discrete time system is $y[n] = \cos(x[n])$

Step 1: Delay input by k units, we get

$$y[n, k] = \cos(x[n - k]) \quad (1.51)$$

Step 2: Replace all n by $n - k$, we get

$$y[n - k] = \cos(x[n - k])$$

Since, $y[n, k] = y[n - k]$, the given system time invariant.

- **Solution 4:** Given discrete time system is $y[n] = |x[n]|$

Step 1: Delay input by k units, we get .

$$y[n, k] = |x[n - k]| \quad (1.52)$$

Step 2: Replace all n by $n - k$, we get

$$y[n - k] = |x[n - k]|$$

Since, $y[n, k] = y[n - k]$, the given system time invariant.

- **Solution 5:** Given discrete time system is $y[n] = x[-n]$

Step 1: Delay input by k units, we get

$$y[n, k] = x[-n - k] \quad (1.53)$$

Step 2: Replace all n by $n - k$, we get

$$\begin{aligned}y[n - k] &= x[-(n - k)] \\&= x[-n + k]\end{aligned}$$

Since, $y[n, k] \neq y[n - k]$, the given system time variant.

- **Solution 6:** Given discrete time system is $y[n] = y[n - 4] + x[n - 4]$

Step 1: Delay input by k units, we get

$$y[n, k] = y[n - 4] + x[n - k - 4] \quad (1.54)$$

Step 2: Replace all n by $n - k$, we get

$$y[n - k] = y[n - k - 4] + x[n - k - 4]$$

Since, $y[n, k] \neq y[n - k]$, the given system time variant.

4. **Causal and non-causal system:** Let $y[n]$ be the output of a discrete-time system for input $x[n]$, then the system is said to be causal if the output at any time instant (n), i.e., $y[n]$ depends only on present and past input, i.e., $(x[n], x[n - 1], x[n - 2], \dots)$. Determine whether the following system is causal or not

(a) $y[n] = x[n]u[n]$

(b) $y[n] = \sum_{k=-\infty}^n x[k]$

- **Solution 1:** Given System is $y[n] = x[n]u[n]$

The output of the system at any time instant n depends only on present input and not on future input. Hence the system is causal.

- **Solution 3:** Given System is $y[n] = \sum_{k=-\infty}^n x[k]$

The given system output at time instant n depends upon:

- $x[-\infty], \dots, x[n-3], x[n-2], x[n-1]$, i.e, past inputs, and
- $x[n]$, i.e., present input.

Hence, the system is causal.

5. Stable and non-stable system: Let $y[n]$ be the output of a discrete-time system for input $x[n]$, then the system is said to be stable, if for bounded input sequence the system gives bounded output sequence, i.e.,

$$|x[n]| < B_x < \infty, \text{ then } |y[n]| < B_y < \infty, \forall n. \quad (1.55)$$

This stable systems are also called Bounded-input Bounded-output (BIBO) systems. In contrast, if for some bounded sequence the system produces unbounded (infinite) sequence, then the system is said to be unstable.

5. A discrete time system has input $x(n)$ and output $y(n)$. The input output relation of the system is given by

$$y(n) = \sum_{k=0}^n x(k)$$

Check whether the system is memory less, time invariant and stable or not?

(2080 Bhadra)

Solution:

Given

$$y(n) = \sum_{k=0}^n x(k)$$

- i) For memory less

A discrete time system is said to be memory less if its output $y[n]$ at $n = n_0$

only depends on input $x[n]$ at $n = n_0$.

In this case $y[n] = x[0] + x[1] + x[2] + \dots + x[n]$

Since the output $y[n]$ at $n = n_0$ depends upon input $x[n]$ at $n \neq n_0$. The system is not memory less.

ii) For time invariant

A discrete time system is said to be time-invariant if a time shift in the input causes an identical time shift in the output.

Here:

$$y(n) = \sum_{k=0}^n x(k)$$

Let us consider a shifted input $x_1[k] = x[n - n_0]$

The output for the shifted input $x_1[n]$ will be:

$$y(n, n_0) = \sum_{k=0}^n x(k - n_0) \quad (1.56)$$

Now, shifting the original output:

$$y(n - n_0) = \sum_{k=0}^{n-n_0} x(k) \quad (1.57)$$

Since both equations (1) and (2) are not equal the system is not time-invariant.

iii) For stability

A discrete time system is said to be stable if every bounded input produces a bounded output.

Here, for all n

$$|x[n]| \leq M < \infty$$

Then

$$|y[n]| = \left| \sum_{k=0}^n x(k) \right| \leq \sum_{k=0}^n M = (n+1)M$$

As $n \rightarrow \infty$, $y[n] \rightarrow \infty$. It grows without bound even though the input is bounded. Thus, the system is not stable.

Determine whether the following system is stable or not

1. $Ax[n] + 10$
2. $y[n] = \cos(x[n])$
3. $y[n] = \sum_{k=-\infty}^n x[k]$

• **Solution 1:** Given System is $y[n] = Ax[n] + 10$

'A' is some arbitrary constant. As long as $x[n]$ is bounded the output is also bounded. Hence the system is stable.

• **Solution 2:** Given System is $y[n] = \cos(x[n])$

Here for every bounded value of $x[n]$ the output $y[n]$, i.e. $\cos(x[n])$ is bounded. Hence, the given system is stable.

• **Solution 3:** Given System is $y[n] = \sum_{k=-\infty}^n x[k]$

let $x[k] = u[k]$, i.e., bounded input signal

Mathematically,

$$u[k] = \begin{cases} 1 & \text{for } k \geq 0, \\ 0 & \text{for } k < 0. \end{cases} \quad (1.58)$$

The system output is:

$$y[n] = \sum_{k=0}^n u[k]$$

$$\text{for } n=3 \implies y[3] = \sum_{k=0}^3 u[k] = u[0] + u[1] + u[2] + u[3] = 4$$

$$\text{for } n=8 \implies y[8] = \sum_{k=0}^8 u[k] = u[0] + u[1] + u[2] + u[3] + u[4] + u[5] + u[6] + u[7] + u[8] = 9$$

As $n \rightarrow \infty$, $y[n] \rightarrow \infty$. Hence the system is unstable.

1.5.2. Interconnection of DTS

There are two basic ways by which DTS can be connected to each other to form larger systems. They are:

1. Cascade or series connection: The series connection of two DTS is shown in Fig. 1.20

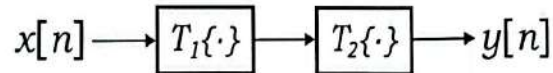


Figure 1.20: Two DTS connected in series.

The output from first DTS is

$$y_1[n] = T_1\{x[n]\},$$

and the output from second DTS is

$$y[n] = T_2\{T_1\{x[n]\}\}.$$

We can combine DTS's T_1 and T_2 to form combined DTS T_s and can write the output of combined system as:

$$y[n] = T_s\{x[n]\} \quad (1.59)$$

where,

$$T_s = T_2 T_1.$$

Note: In general, the order of operation of DTS T_1 and T_2 is important.

Therefore, in general $T_1 T_2 \neq T_2 T_1$

2. Parallel connection: The parallel connection of two DTS is shown in Fig.

1.21

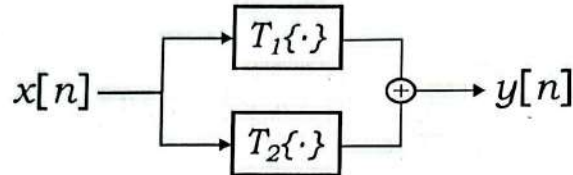


Figure 1.21: Two DTS connected in parallel.

The output from first DTS is

$$y_1[n] = T_1\{x[n]\},$$

and the output from second DTS is

$$y_2[n] = T_2\{x[n]\}.$$

The combined output is given by:

$$y[n] = T_1\{x[n]\} + T_2\{x[n]\}.$$

We can combine DTS's T_1 and T_2 to form combined DTS T_p and can

write the output of combined system as:

$$y[n] = T_p\{x[n]\}, \quad (1.60)$$

where,

$$T_p = T_1 + T_2.$$

1.5.3. Response of a discrete-time system

The response of a discrete-time system is characterized by the type of input sequence.

1. **Impulse response:** The response of a discrete time system when the input sequence is an impulse or unit sample sequence, i.e., $\delta[n]$. The impulse response is denoted by $h[n]$.
2. **Unit response:** The response of a discrete time system when the input sequence is a unit step sequence. The unit step response is denoted by $s[n]$.

1.6. Linear-Time Invariant DTS

A discrete time system that is both **linear** and **time-invariant** is called **Linear Time-Invariant (LTI)** system. An LTI system is completely specified by its impulse response. The impulse response of LTI system is the response of the system when the input is an impulse sequence, i.e., $\delta[n]$. The impulse response is denoted by $h[n]$. Now, we derive the output of the LTI system to arbitrary input $x[n]$. Recall that any arbitrary sequence can be repre-

sented as scaled and shifted impulses. Therefore, any arbitrary sequence can be written as:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta(n-k) \quad (1.61)$$

let $y[n]$ be the output of LTI system for the above sequence $x[n]$. Then,

$$\begin{aligned} y[n] &= T\{x[n]\} \\ &= T\left\{\sum_{k=-\infty}^{\infty} x[k]\delta(n-k)\right\} \\ &= \sum_{k=-\infty}^{\infty} x[k]T\{\delta(n-k)\} \text{ from Linearity property of LTI} \\ &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \text{ from Time invariant property of LTI} \end{aligned}$$

Therefore the output $y[n]$ of LTI system for any input $x[n]$ is given as:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h(n-k) \quad (1.62)$$

The expression in equation 1.62 is called **Convolution sum or linear convolution**. In short it is written as:

$$y[n] = x[n] * h[n] \quad (1.63)$$

1.6.1. Properties of LTI system

The output of LTI system is given by convolution sum. Therefore general properties of the class of LTI systems can be found by considering properties of the convolution operation.

1. Commutative: The convolution sum satisfies the Commutative property

so does any LTI system. The Commutative property of LTI system is:

$$x[n] * h[n] = h[n] * x[n]$$

proof:

$$\begin{aligned}\text{L.H.S} &= x[n] * h[n] \\ &= \sum_{k=-\infty}^{\infty} x[k]h(n-k) \\ \text{let } m=n-k, \\ &= \sum_{m=-\infty}^{-\infty} x[n-m]h[m] \\ &= \sum_{m=-\infty}^{\infty} h[m]x[n-m] \\ &= h[n] * x[n] \\ &= \text{R.H.S}\end{aligned}$$

2. Distributive: Any LTI system satisfies distributive law:

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

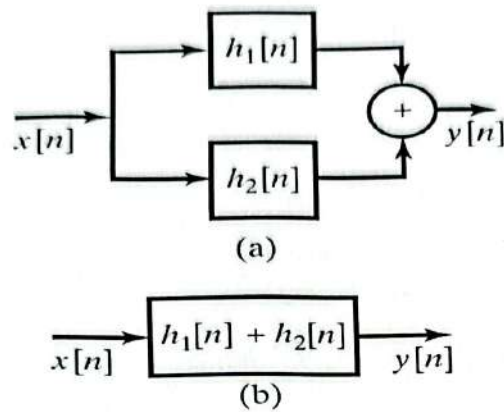


Figure 1.22: a) Parallel combination of LTI systems. (b) An equivalent system using distributive law.

proof:

$$\begin{aligned}
 \text{L.H.S} &= x[n] * (h_1[n] + h_2[n]) \\
 &= \sum_{k=-\infty}^{\infty} (x[k](h_1[n-k] + h_2[n-k])) \\
 &= \sum_{k=-\infty}^{\infty} (x[k]h_1[n-k] + x[k]h_2[n-k]) \\
 &= \sum_{k=-\infty}^{\infty} x[k]h_1[n-k] + \sum_{k=-\infty}^{\infty} x[k]h_2[n-k] \\
 &= x[n] * h_1[n] + x[n] * h_2[n] \\
 &= \text{R.H.S}
 \end{aligned}$$

Distributive law is associated with parallel connection of LTI system as shown in Fig. 1.22

3. Associative:

$$(x[n] * (h_1[n] * h_2[n])) = x[n] * (h_1[n] * h_2[n])$$

Associative law is associated with cascade or series connection of LTI system as shown in Fig. 1.23

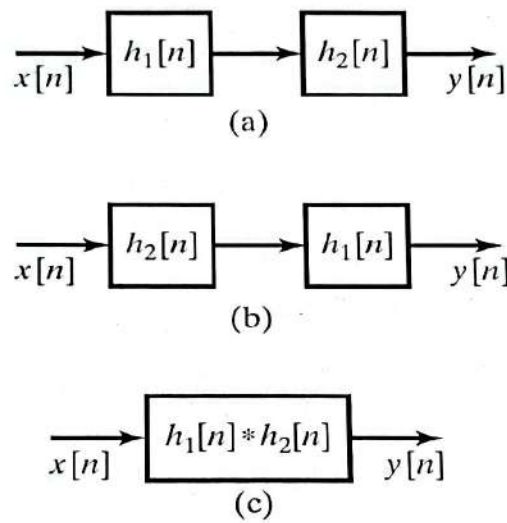


Figure 1.23: a) Series combination of LTI systems. (b) An equivalent system using associative law.

4. Causality: Derive the condition for the LTI system to be causal. We know, the output of LTI system is

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

at any n say $n = n_0$

$$\begin{aligned} y[n_0] &= \sum_{k=-\infty}^{\infty} x[k]h[n_0 - k] \\ &= \sum_{k=-\infty}^{-1} x[k]h[n_0 - k] + \sum_{k=0}^{\infty} x[k]h[n_0 - k] \end{aligned}$$

for causality,

$$\sum_{k=-\infty}^{-1} x[k]h[n_0 - k] = 0 \quad (1.64)$$

for equation (1.64) to hold true

$$h[n_0] = 0 \text{ for } n_0 < 0, \text{ or}$$

$$h[n] = 0 \text{ for } n < 0$$

Note: The output of the linear convolution formula for causal LTI system can be written as:

$$y[n] = \sum_{k=0}^{\infty} x[k]h[n - k] \quad (1.65)$$

or,

$$y[n] = \sum_{k=0}^{\infty} h[k]x[n - k] \quad (1.66)$$

5. Stable: Condition for the stability of LTI system? We know, the output of LTI system is

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k]$$

or,

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n - k] \quad (1.67)$$

Let the input sequence $x[n]$ is bounded, i.e.,

$$|x[n]| \leq M_x < \infty \quad (1.68)$$

where,

M_x is some finite number.

Taking absolute value on both side of 1.67, we get,

$$\begin{aligned} |y[n]| &= \left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right| \\ |y[n]| &\leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]| \end{aligned} \quad (1.69)$$

$$\left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right| \leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]|$$

From equation 1.68, we have

$$|x[n]| < M_x < \infty$$

Time shifting does not change amplitude. Therefore,

$$|x[n-k]| \leq M_x < \infty$$

Now, equation 1.69 can be written as,

$$|y[n]| \leq \sum_{k=-\infty}^{\infty} |h[k]| M_x$$

$$|y[n]| \leq M_x \sum_{k=-\infty}^{\infty} |h[k]| \quad (1.70)$$

For equation 1.70 to be finite,

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty \quad (1.71)$$

Therefore, an LTI system is stable if the impulse response is absolutely summable.

Example 1: Find the output of an LTI system with input sequence $x[n] = \{1, 3, 2, 0.5\}$ and impulse response $h[n] = \{2, 3, -4, 5\}$

Solution: We know that the output of LTI system is given by

$$\begin{aligned} y[n] &= x[n] * h[n] \\ &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\ &= \dots + x[-2]h[n+2] + x[-1]h[n+1] + x[0]h[n] + x[1]h[n-1] + \dots \end{aligned}$$

for $n = -2$ above equation becomes

$$\begin{aligned} y[-2] &= x[-2]h[-2+2] + x[-1]h[-2+1] + x[0]h[-2] + x[1]h[-2-1] \\ &= x[-2]h[0] + x[-1]h[-1] + x[0]h[-2] + x[1]h[-3] \\ &= 1(2) + 3(0) + 2(0) + 0.5(0) \\ &= 2 \end{aligned}$$

for $n = -1$ above equation becomes

$$\begin{aligned}y[-1] &= x[-2]h[-1+2] + x[-1]h[-1+1] + x[0]h[-1] + x[1]h[-1-1] \\&= x[-2]h[1] + x[-1]h[0] + x[0]h[-1] + x[1]h[-2] \\&= 1(3) + 3(2) + 2(0) + 0.5(0) \\&= 9\end{aligned}$$

for $n = 0$ above equation becomes

$$\begin{aligned}y[0] &= x[-2]h[0+2] + x[-1]h[0+1] + x[0]h[0] + x[1]h[0-1] \\&= x[-2]h[2] + x[-1]h[1] + x[0]h[0] + x[1]h[-1] \\&= 1(-4) + 3(3) + 2(2) + 0.5(0) \\&= 9\end{aligned}$$

for $n = 1$ above equation becomes

$$\begin{aligned}y[1] &= x[-2]h[1+2] + x[-1]h[1+1] + x[0]h[1] + x[1]h[1-1] \\&= x[-2]h[3] + x[-1]h[2] + x[0]h[1] + x[1]h[0] \\&= 1(5) + 3(-4) + 2(3) + 0.5(2) \\&= 0\end{aligned}$$

for $n = 2$ above equation becomes

$$\begin{aligned}y[2] &= x[-2]h[2+2] + x[-1]h[2+1] + x[0]h[2] + x[1]h[2-1] \\&= x[-2]h[4] + x[-1]h[3] + x[0]h[2] + x[1]h[1] \\&= 1(0) + 3(5) + 2(-4) + 0.5(3) \\&= 8.5\end{aligned}$$

for $n = 3$ above equation becomes

$$\begin{aligned}y[3] &= x[-2]h[3+2] + x[-1]h[3+1] + x[0]h[3] + x[1]h[3-1] \\&= x[-2]h[5] + x[-1]h[4] + x[0]h[3] + x[1]h[2] \\&= 1(0) + 3(0) + 2(5) + 0.5(-4) \\&= 8\end{aligned}$$

for $n = 4$ above equation becomes

$$\begin{aligned}y[4] &= x[-2]h[4+2] + x[-1]h[4+1] + x[0]h[4] + x[1]h[4-1] \\&= x[-2]h[6] + x[-1]h[5] + x[0]h[4] + x[1]h[3] \\&= 1(0) + 3(0) + 2(0) + 0.5(5) \\&= 2.5\end{aligned}$$

Therefore, $y[n] = \{2, 9, 9, 0, 8.5, 8, 2.5\}$

2. Find the output of LTI system having impulse response $h[n] = u[n] - u[n-4]$

and input signal $x[n] = \frac{1}{2}^n u[n]$. (2078 Bhadra)

Solution:

Given

$x[n]/$ $h[n]$	$x[-2] =$ 1	$x[-1] =$ 3	$x[0] =$ 2	$x[1] =$ 0.5
$h[0] = 2$	2	6	4	1
$h[1] = 3$	3	9	6	1.5
$h[2] = -4$	-4	-12	-8	-2
$h[3] = 5$	5	15	10	2.5

$x[n]/$ $h[n]$	$x[-2] =$ 1	$x[-1] =$ 3	$x[0] =$ 2	$x[1] =$ 0.5	$\Sigma = 2$
$h[0] = 2$	2	6	4	1	$\Sigma = 9$
$h[1] = 3$	3	9	6	1.5	$\Sigma = 9$
$h[2] = -4$	-4	-12	-8	-2	$\Sigma = 0$
$h[3] = 5$	5	15	10	2.5	$\Sigma = 8.5$
					$\Sigma = 8$
					$\Sigma = 2.5$

Figure 1.24: Checking answer of linear convolution

$$x[n] = \frac{1}{2} u[n]$$

and

$$h[n] = u[n] - u[n - 4]$$

Here,

$$h[n] = \begin{cases} 1, & \text{for } 0 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

We know that the output of the LTI system is given by:

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

For $n < 0$, $y[n] = 0$

Since

$$x[k] = \frac{1}{2}^k u[k]$$

$x[k] \neq 0$ for $k \geq 0$.

Similarly, $h[n] \neq 0$ for $0 \leq n - k \leq 3$ which means the limit becomes

$$n - 3 \leq k \leq n$$

Thus the summation limit becomes $\max(0, n - 3) \leq k \leq n$

Now for $0 \leq n \leq 3$, the summation range is $0 \leq k \leq n$

$$\begin{aligned} y[n] &= \sum_{k=0}^n \left(\frac{1}{2}\right)^k \\ &= \left(\frac{1}{2}\right)^0 \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}} \end{aligned}$$

which is in the form $\frac{a(1-r^N)}{1-r}$

$$= 2\left(1 - \left(\frac{1}{2}\right)^{n+1}\right)$$

For $n > 3$, the summation range is $n - 3 \leq k \leq n$

$$\begin{aligned}
 y[n] &= \sum_{k=n-3}^n \left(\frac{1}{2}\right)^k \\
 &= \left(\frac{1}{2}\right)^{n-3} \sum_{j=0}^3 \left(\frac{1}{2}\right)^j \\
 &= \left(\frac{1}{2}\right)^{n-3} \frac{1 - \left(\frac{1}{2}\right)^4}{1 - \frac{1}{2}} \\
 &= \frac{15}{8} \left(\frac{1}{2}\right)^{n-3}
 \end{aligned}$$

Therefore the output of the LTI system will be:

$$y[n] = \begin{cases} 0, & \text{for } n < 0 \\ 2\left(1 - \left(\frac{1}{2}\right)^{n+1}\right), & \text{for } 0 \leq n \leq 3 \\ \frac{15}{8} \left(\frac{1}{2}\right)^{n-3}, & \text{for } n > 3 \end{cases}$$

4. Find the output of LTI system having input signal $x[n] = \delta[n] + 2\delta[n - 1] - \delta[n - 3]$ and $h[n] = 2\delta[n + 1] + 2\delta[n - 1]$. (2079 Bhadra, 2081 Bhadra)

Solution:

Given

$$\begin{aligned}
 x[n] &= \delta[n] + 2\delta[n - 1] - \delta[n - 3] \\
 &= \{1, 2, 0, -1\} \text{ for } n = 0, 1, 2, 3
 \end{aligned}$$

and

$$h[n] = 2\delta[n + 1] + 2\delta[n - 1]$$

$$= \{2, 0, 2\} \text{ for } n = -1, 0, 1$$

The output of the LTI system is given by:

$$y[n] = x[n] * h[n]$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

In this case

$$\begin{aligned} y[n] &= \sum_{k=0}^3 x[k]h[n-k] \\ &= x[0]h[n] + x[1]h[n-1] + x[2]h[n-2] + x[3]h[n-3] \end{aligned}$$

The length of the output $y[n] = M + N - 1 = 4 + 3 - 1 = 6$

where M is the length of $x[n]$ and N is the length of $h[n]$

Also,

The starting index is the sum of minimums of signals $x[n]$ and $h[n]$. So in this case $\min(x[n], h[n]) = \min(0, -1) = -1$

Thus the output sequence will be as follows:

$$\begin{aligned} y[-1] &= x[0]h[-1] + x[1]h[-2] + x[2]h[-3] + x[3]h[-4] \\ &= 1 * 2 + 2 * 0 + 0 * 0 - 1 * 0 = 2 \end{aligned}$$

Similarly we get,

$$y[0] = 1 * 0 + 2 * 2 + 0 * 0 - 1 * 0 = 4$$

$$y[1] = 1 * 2 + 2 * 0 + 0 * 2 - 1 * 0 = 2$$

$$y[2] = 1 * 0 + 2 * 2 + 0 * 0 - 1 * 2 = 2$$

$$y[3] = 1 * 0 + 2 * 0 + 0 * 0 - 1 * 0 = 0$$

$$y[4] = 1 * 0 + 2 * 0 + 0 * 0 - 1 * 2 = -2$$

∴ The output sequence $y[n] = \{2, 4, 2, 2, 0, 2\}$.

Practice: Find the output of an LTI system with input sequence $x[n] = \{1, 3, 4, -5\}$ and impulse response $h[n] = \{0.5, 2, -3, 5, 4\}$.

Example: Find the output of LTI system with input sequence

$$x[n] = \left(\frac{1}{3}\right)^n u[n]$$

and impulse response

$$h[n] = u[n + 2] - u[n - 2]$$

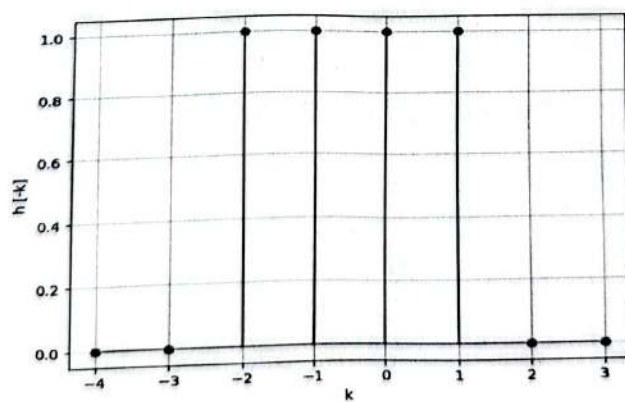
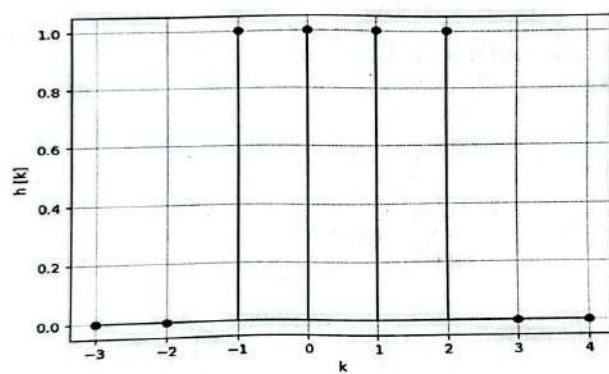
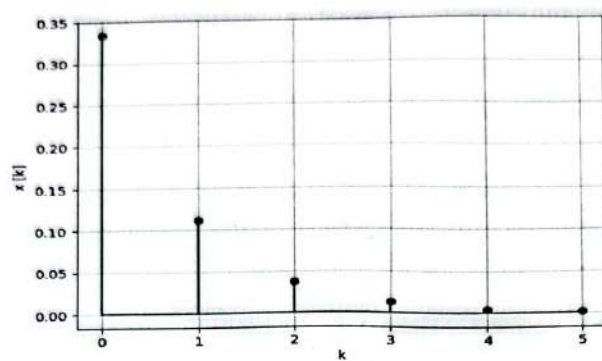
Solution: We know that the output of LTI system is given by

$$\begin{aligned} y[n] &= x[n] * h[n] \\ &= \sum_{k=-\infty}^{\infty} x[k]h[n - k] \end{aligned}$$

Plotting the given signals to determine the range of the output sequence we get,

- for $n < -2$, $x[k]h[n - k] = 0$, Therefore, $y[n] = 0$ for $-2 \leq n \leq -2$,

Therefore, $y[n] = 0$



1.7. Frequency Domain Representation of Sequence

Until now we have represented sequence in time domain. To represent sequence in frequency domain we use Fourier analysis: Fourier series (for periodic sequence) and Fourier Transform (for aperiodic sequence).

1.7.1. Discrete Time Fourier Series of periodic signal

1. **Discrete Time Fourier Series (DTFS):** Given an N -periodic sequence $x[n]$, i.e., $x[n + N] = x[n]$, for all n . The complex exponential sequence is $e^{j\omega_0 n}$, its harmonically related complex exponentials are:

$$e^{j0\omega_0 n}, e^{j1\omega_0 n}, e^{j2\omega_0 n}, \dots, e^{j(N-1)\omega_0 n}, \text{ or in short } e^{jk\omega_0 n}, k = 0, 1, 2, \dots, N - 1$$

Fourier series states that $x[n]$, which is N -periodic can be represented as linear combination of N harmonically related complex exponentials. Mathematically,

$$x[n] = \sum_{k=0}^{N-1} X[k] e^{jk\omega_0 n} \quad (1.72)$$

where,

$$\omega_0 = \frac{2\pi}{N}$$

$X[k]$ are the Fourier coefficients, and is given by

$$X[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n}, \quad k = 0, 1, 2, \dots, N-1 \quad (1.73)$$

Equation 1.72 is called Discrete-Time Fourier Series (DTFS).

Notes:

- (a) We only need N complex exponentials to represent a DT periodic signal with period N . Indeed, there are only N distinct complex exponentials with frequencies that are integer multiples of $\frac{2\pi}{N}$. The Fourier coefficients $X[k], k = 0, 1, \dots, N-1$ are complex valued.
- (b) Plot of $|X[k]|$ versus k is called the **magnitude spectrum** of $x[n]$, and it gives the information about the frequency content of the sequence $x[n]$.
- (c) Plot of $\angle X[k]$ versus k is called the **phase spectrum** of $x[n]$, and it gives the information about the phase content of the sequence $x[n]$.
- (d) Plot of $|X[k]|^2$ versus k is called the **power density spectrum** of $x[n]$, and it gives the information about the distribution of power as a function of frequency of the sequence $x[n]$.
- (e) The fourier coeffecients are also periodic in N , i.e., $X[k + N] = X[k]$, i.e., the spectrum of N -periodic sequence $x[n]$, is also N -periodic.

2. **Parsevals power theorem** (Average power of N-periodic sequence):

The average power of N-periodic sequence is given by:

$$P_{avg} = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2 \quad (1.74)$$

Fourier series of $x[n]$:

$$x[n] = \sum_{k=0}^{N-1} X[k] e^{jk\omega_0 n} \quad (1.75)$$

Fourier series of $x^*[n]$:

$$x^*[n] = \sum_{k=0}^{N-1} X^*[k] e^{-jk\omega_0 n} \quad (1.76)$$

Equation 1.74 can be written as:

$$P_{avg} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] x^*[n]$$

substitute equation 1.76 in 1.74, we get

$$\begin{aligned} &= \frac{1}{N} \sum_{n=0}^{N-1} x[n] \sum_{k=0}^{N-1} X^*[k] e^{-jk\omega_0 n} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X^*[k] \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n} \\ &= \sum_{k=0}^{N-1} X^*[k] X[k] \\ &= \sum_{k=0}^{N-1} |X[k]|^2 \end{aligned}$$

Therefore,

$$P_{avg} = \sum_{k=0}^{N-1} |X[k]|^2 \quad (1.77)$$

Equation 1.77 is called **Parseval's power theorem**

3. **Dirichlet condition:** Dirichlet condition provides the condition for the existence of Fourier series of N-periodic sequence $x[n]$:

- (a) The N-periodic sequence $x[n]$ must be absolutely summable.
- (b) The N-periodic sequence $x[n]$ must have finite number of Maxima and Minima.
- (c) The N-periodic sequence $x[n]$ must have finite number of discontinuities.

Properties of DTFS: Let $x[n]$ and $y[n]$ are both N periodic. Fourier series exhibits various properties listed below:

- **Linearity:** Fourier series satisfies superposition and homogeneity principle.

If $x_1[n] \leftrightarrow X[k_1]$ and $x_2[n] \leftrightarrow X[k_2]$, then

$$a_1 x_1[n] + a_2 x_2[n] = a_1 X[k_1] + a_2 X[k_2] \quad (1.78)$$

- **Shifting:** Shifting in time equals a phase shift of Fourier coefficients. If

$x_1[n] \leftrightarrow X[k_1]$, then

$$x_1[n - n_0] \leftrightarrow X[k_1] e^{-jk\omega_0 n_0} \quad (1.79)$$

- **Parseval's theorem:**

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 \quad (1.80)$$

1.7.2. Discrete Time Fourier Transform (DTFT) of aperiodic signal

The Fourier transform of aperiodic discrete time sequence $x[n]$ is given by

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad (1.81)$$

- $X(\omega)$ provides the frequency content of the aperiodic sequence $x[n]$.
- $X(\omega)$ is a continuous function of frequency ω .

Notes:

1. $X(\omega)$ is periodic with period 2π . This is due to the fact that the frequency range of any discrete-time sequence is limited to $(-\pi, \pi)$ or $(0, 2\pi)$.
2. Plot of $|X(\omega)|$ versus ω is called the **magnitude spectrum** of $x[n]$, and it gives the information about the frequency content of the sequence $x[n]$.
3. Plot of $\angle X(\omega)$ versus ω is called the **phase spectrum** of $x[n]$, and it gives the information about the phase content of the sequence $x[n]$.
4. Plot of $|X(\omega)|^2$ versus ω is called the **energy density spectrum** of $x[n]$, and it gives the information about the distribution of energy as a function of frequency of the sequence $x[n]$.

Expression for $x[n]$:

Multiply both side of equation 1.81 by $e^{j\omega m}$ and integrate from $(-\pi, \pi)$

$$\int_{-\pi}^{\pi} X(\omega)e^{j\omega m}d\omega = \int_{-\pi}^{\pi} \left[\sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \right] e^{j\omega m} d\omega$$

interchanging the order of summation and integration, we get

$$\int_{-\pi}^{\pi} X(\omega) e^{j\omega m} d\omega = \sum_{n=-\infty}^{\infty} \int_{-\pi}^{\pi} x[n] e^{j\omega(m-n)} d\omega \quad (1.82)$$

we know,

$$\int_{-\pi}^{\pi} e^{j\omega(m-n)} d\omega = \begin{cases} 2\pi, & m = n \\ 0, & m \neq n \end{cases}$$

consequently,

$$\sum_{n=-\infty}^{\infty} \int_{-\pi}^{\pi} x[n] e^{j\omega(m-n)} d\omega = \begin{cases} 2\pi x[m], & m = n \\ 0, & m \neq n \end{cases} \quad (1.83)$$

from equation 1.82 and 1.83, we get

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X[\omega] d\omega \quad (1.84)$$

Table ?? shows the synthesis and analysis equation for the Fourier Transform.

Table ?? shows the short hand notation of DTFT:

1.7.2.1. Condition for the existence of DTFT:

- The discrete-time sequence $x[n]$ should be absolutely summable.

1.7.2.2. Discrete-Time Fourier Transform (DTFT) properties:

The properties of DTFT are useful for reducing the complexity of frequency analysis problem. Below we list some of the properties of DTFT:

1. Periodicity: The periodicity property of DTFT states that if

$$x[n] \xleftrightarrow{FT} X(\omega)$$

then,

$$X[\omega + 2\pi] \xleftrightarrow{FT} X(\omega) \quad (1.85)$$

2. Linearity: The linearity property of DTFT states that if

$$x_1[n] \xleftrightarrow{FT} X_1(\omega)$$

$$x_2[n] \xleftrightarrow{FT} X_2(\omega)$$

then,

$$a_1x_1[n] + a_2x_2[n] \xleftrightarrow{FT} a_1X_1[\omega] + a_2X_2(\omega) \quad (1.86)$$

3. Time shifting: The time shifting property of DTFT states that if

$$x[n] \xleftrightarrow{FT} X(\omega)$$

then,

$$x[n - n_0] \xleftrightarrow{FT} X(\omega)e^{-j\omega n_0} \quad (1.87)$$

4. Frequency shifting: The frequency shifting property of DTFT states that

if

$$x[n] \xleftrightarrow{FT} X(\omega)$$

then,

$$x[n]e^{j\omega_0 n} \xleftrightarrow{FT} X(\omega - \omega_0) \quad (1.88)$$

5. **Modulation:** The modulation property of DTFT states that if

$$x[n] \xleftrightarrow{FT} X(\omega)$$

then,

$$x[n] \cos \omega_0 n \xleftrightarrow{FT} \frac{1}{2}(X((\omega + \omega_0) + (\omega - \omega_0))) \quad (1.89)$$

6. **Time reversal:** The time reversal property of DTFT states that if

$$x[n] \xleftrightarrow{FT} X(\omega)$$

then,

$$x[-n] \xleftrightarrow{FT} X(-\omega) \quad (1.90)$$

$$x^*[-n] \xleftrightarrow{FT} X^*(\omega) \quad (1.91)$$

7. **Convolution:** The convolution property of DTFT states that if

$$x[n] \xleftrightarrow{FT} X[\omega]$$

$$y[n] \xleftrightarrow{FT} Y[\omega]$$

then,

$$x[n] * y[n] \xleftrightarrow{FT} X(\omega)Y(\omega) \quad (1.92)$$

8. **Multiplication:** The multiplication property of DTFT states that if

$$x[n] \xleftrightarrow{FT} X(\omega)$$

$$y[n] \xleftrightarrow{FT} Y(\omega)$$

then,

$$x[n]y[n] \xleftrightarrow{FT} \frac{1}{2\pi} X(\omega) * Y(\omega) \quad (1.93)$$

9. **Parseval's Energy Theorem:** The Parseval's energy theorem states that if

$$x[n] \xleftrightarrow{FT} X(\omega)$$

then,

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 \xleftrightarrow{FT} \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega \quad (1.94)$$

1. Find the discrete-time Fourier transform $x[n] = \{1, 3, -2, 5, 2\}$.

Solution: We know DTFT is given as

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

$$\begin{aligned} X(\omega) &= \sum_{n=0}^3 x[n]e^{-j\omega n} \\ &= x[0]e^{-j\omega 0} + x[1]e^{-j\omega 1} + x[2]e^{-j\omega 2} + x[3]e^{-j\omega 3} + x[4]e^{-j\omega 4} \\ &= 1e^{-j\omega 0} + 3e^{-j\omega 1} + -2e^{-j\omega 2} + 5e^{-j\omega 3} + 2e^{-j\omega 4} \end{aligned}$$

2. Find the discrete-time Fourier transform of the sequence

$$x[n] = u[n] = \begin{cases} 1 & \text{for } n \geq 0, \\ 0 & \text{for } n < 0. \end{cases} \quad (1.95)$$

Solution: We know DTFT is given as

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad (1.96)$$

From equation 1.95 and 1.96 we get,

$$\begin{aligned} X(\omega) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} u[n]e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} 1e^{-j\omega n} \\ &= \frac{1}{1 - e^{-j\omega}} \end{aligned}$$

1.7.3. Frequency response of LTI system

To be studied in Chapter 3.

1.7.4. Sampling of Continuous time signal

Sampling is the process of converting continuous time signal to discrete time sequence. In general a discrete time sequence is obtained from periodic sampling of continuous time signal. Sampling is done using continuous to discrete time (C/D) converter. Fig 1.25 shows block diagram

of an ideal C/D converter. let $x_c(t)$ be a continuous-time signal then the

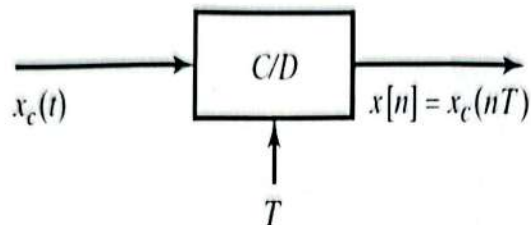


Figure 1.25: Block diagram of C/D converter

output of C/D converter is:

$$x[n] = x_c(nT) \quad (1.97)$$

where,

T is the sampling period,

and its reciprocal, $f_s = 1/T$, is the sampling frequency, in samples per second.

Note :

1. We can also express the sampling frequency as $\omega_s = \frac{2\pi}{T}$ in radians per second.
2. Sampling representations rely only on the assumption of a bandlimited Fourier transform.

Mathematically, sampling process is carried out in two stages as shown in Fig. 1.26. The stages consist of an impulse train modulator, followed by conversion of the impulse train to a sequence. The periodic impulse train is Frequency domain representation of sampling process:

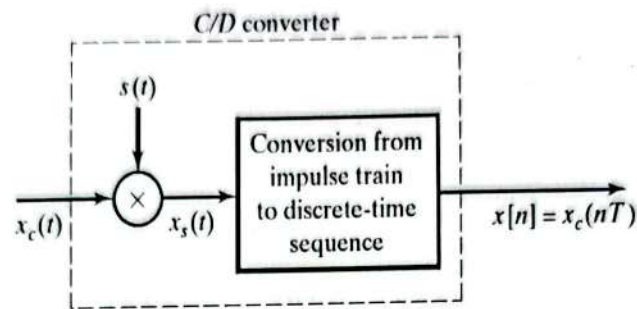


Figure 1.26: Two stages of sampling process.

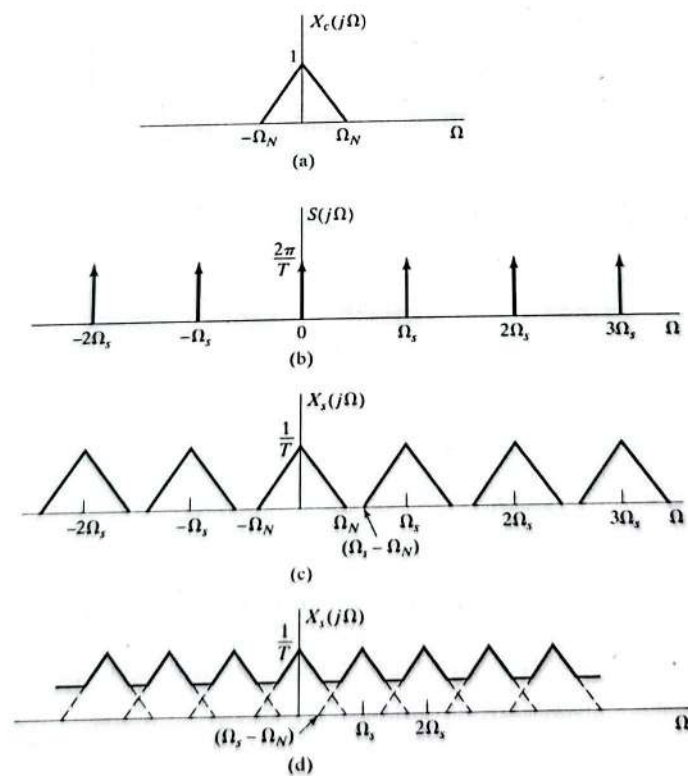


Figure 1.27: Spectral properties of sampled signal.